

# Hunner-lesion interstitial cystitis after the audit: a preregistered net-of-inflammation residual program with transparent negative results

MEGA-PROGRAM-27, interstitial cystitis lane (builder branch builder-25-ic)

26 September 2026 - working disease manuscript; not yet complete

## Abstract

Interstitial cystitis/bladder pain syndrome with Hunner lesions (HIC) is a distinct inflammatory bladder disease whose identification still rests on cystoscopy, not on a molecular marker. This report does four things, in order, and keeps every negative on the record. First, a duplicate-sample and mislabel audit shows the historical top-50 IC expression panel does not survive correction: in the post-audit two-source replication the old and new top-50 lists share zero genes, no gene reaches BH  $q < .05$ , and the registered sign-agreement threshold fails (43/50,  $p = .0469$  against a .025 bar). Second, we acquire and individually verify 98 new accession records, bringing the lane to 153 accession-backed records with honest nesting labels; the statistical evidence behind them rests on 25 paired HIC patients, 13 BCG samples, 13 urine-sediment samples and 20 EV-miRNA subjects, and the record count is an audit figure, not a replication unit. Third, we preregister a module-level test on the 2025 paired Hunner-lesion RNA-seq cohort (GSE238208; 25 patients, lesion and non-lesion biopsies, plus 13 BCG inflammatory-cystitis samples): the lesion program replicates across patients (19/25,  $p = .0073$ ) but is shared with BCG inflammation, so the Hunner-specific gate fails. Fourth, a locked amendment tests the net-of-BCG residual: a BCG-adjusted, G2/M-associated transcriptional residual is reproducible across held-out patients within this cohort (18/25,  $p = .0216$ ). The locked specificity bound is very wide (compatible with no specificity and with strong specificity), and two pre-specified exploratory urinary transfers are inconclusive - one small directionally consistent signal carried substantially by a single high sample, and one non-significant EV-miRNA test. Forty-three external scientific services annotate the program (41 under the most conservative recount). No clinical test, named discovery, or benchmark break is claimed; this is a working research manuscript, not a diagnostic claim.

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# 1 Scope, verdict and reproducibility

This document is the disease-level manuscript for the interstitial-cystitis lane of MEGA-PROGRAM-27. It is written while the project gates are only partly met, and says so. Current gate state: the 120 accession-record floor is met (153 records, Table 1); the 40 genuinely-used external service floor is met (43 counted, 41 most-conservative, Appendix E); the substantive-page floor for this disease paper is not met and is stated honestly here: at this revision the manuscript renders 43 pages, of which the research-body audit (pages carrying at least 300 non-heading words) counts 10 of the 50 required - the remainder of the page count is locked result tables, gene-set appendices, per-fold values and the service ledger, which are data, not prose. Per the lane’s substance-first rule, no padding was added to reach the floor; the shortfall is recorded rather than disguised; a positive preregistered replication exists for the residual program (Gate A2.1) but no published-comparator benchmark break and no named new discovery is claimed. Every analysis threshold in this manuscript was committed to the repository before the corresponding test was run; preregistrations are `prereg/IC_P48_preregistration.md` and `prereg/IC_P48_A2_amendment.md`, and every test cites the JSON artifact its numbers come from.

The lane began this phase with 55 IC-tagged records in the shared global manifest (1,508 records across diseases); the P45–P47 acquisition added 98 distinct accession records with no collision against any existing entry, moving the lane to 153 and the global manifest to 1,606. Record units are accession records with verified roles, not independent studies, cohorts or people.

## 2 Disease background and why the old panel had to be re-audited

Interstitial cystitis/bladder pain syndrome (IC/BPS) splits clinically into Hunner-lesion disease (HIC), an inflammatory phenotype with ulcerative lesions visible at cystoscopy, and non-lesion bladder pain syndrome (BPS), which shares pain but not the lesion biology [11, 12]. BCG intravesical therapy for bladder cancer produces an iatrogenic inflammatory cystitis [15] that is the most important molecular comparator: any “Hunner signature” that is equally present in BCG cystitis is a general severe-inflammation signature, not a Hunner marker. The historical in-program IC panel descended from two array series (GSE621, GSE11783). The program-wide audit found two IC-derived mock-APF samples in GSE621 had been treated as normal controls, and that GSE11783’s ten IC libraries are nested in five patients, so library-level inference was not patient-level. The post-audit sensitivity (Section 4) is therefore the correct statement of the historical panel: it is negative.

## 3 Data acquisition and provenance

Each GSM record used in this lane was fetched individually (SOFT text retained in `sources/`), mapped to its matrix column by deposited title and description, and labeled with its role, including honest nesting (for example, “not independent cohort/participant” where libraries repeat a person). The P45 acquisition (GSE238208) contributes 64 sample records: 25 paired Hunner lesion and non-lesion biopsies and 13 BCG samples across 38 people. P46 (GSE196156) contributes 21 urinary extracellular-vesicle miRNA records (8 cystitis, 2 BPS, 10 controls, one platform). P47 (GSE11839) contributes 13 cultured-urothelium records (3 IC, 3 controls, two media). Together with the re-audited historical records and the individually checked GSE57560 GSMs, the lane holds 153 accession-backed records (145 sample records in Table 1 plus eight series-level records). Duplicate accessions against the global manifest were checked before appending; the manifest’s unit is a unique accession-backed record, not an independent cohort. The ledger arithmetic is: 55 records already credited to this lane (across GSE621, GSE11783, GSE28242 and GSE57560) plus 98 new record units (64 + 21 + 13 from P45, P46 and P47) equals 153; of these, 145 are sample-level records and eight are series-level records. The 120-record floor is therefore met with a margin of 33, and every record carries the people-count caveats of its source (4 and the provenance subsections) so that record units are never read as independent studies.

### 3.1 Acquisition audit trail

The acquisition audit (`results/ic_p45_p47_acquisition_audit.json`) records, per series, the exact verification performed before any record entered the manifest. For GSE238208: series SOFT fetched; each of 63 GSM sample sections split and stored individually; deposited `Sample_description` PR identifiers matched 63/63 to named author-FPKM columns; all matched columns fully finite across 27,339 gene rows. For GSE196156: all 20 GSMs present as series-matrix columns, each carrying 464 finite miRNA measurements of 2,632 deposited rows, with the remaining rows empty across all samples - a detection-gated miRNA output, uniform across columns, which is why the measured set is defined as the 464 fully observed rows rather than imputed. For GSE11839: all 12 GSMs present as matrix columns, fully finite across 54,675 rows. New records were checked against the live global manifest for collisions before appending, and the four already-credited series (GSE621, GSE11783, GSE28242, GSE57560) were explicitly not re-counted.

### 3.2 Per-dataset provenance and verification

**GSE238208** (P45; RNA-seq, GPL24676 NovaSeq 6000; iScience 2023 [1]). Twenty-five Hunner-type IC patients contribute paired lesion and non-lesion bladder biopsies (50 GSM), and thirteen BCG-cystitis biopsies form the inflammatory comparator block; 38 people, 63 GSM total. Each of the 63 sample SOFT sections was split and stored individually; the deposited `Sample_description` PR identifiers were matched 63/63 to the named columns of the author FPKM matrix, and every matched column is fully finite across all 27,339 gene rows. This is the primary inference dataset; the unit of inference is the patient, never the library.

**GSE196156** (P46; urinary extracellular-vesicle miRNA-seq, GPL29077). Eight cystitis, two BPS and ten control urine-EV profiles (20 GSM). The series matrix holds 2,632 miRNA rows, of which 464 are fully measured across all 20 samples; rows with missing calls are excluded by the locked definition of the measured set, and the group order was verified against the individually stored sample records. The two BPS profiles are retained for description only.

**GSE11839** (P47; cultured urothelium array; J. Urology 2008 [2]). Three IC and three control urothelial cultures under two media (13 GSM). Used for acquisition completeness and context; cultured cells are not a lesion surrogate, and no gate is run on this series.

**GSE28242** (urine sediment array; J. Urology 2012 [3]). Five control, five without-lesion and three with-lesion urine-sediment profiles (13 GSM), individually re-checked against their sample records. Hosts the exploratory A2.3a transfer; the with-lesion arm is three donors.

**GSE57560** (bladder biopsies; J. Urology 2014 [4]). All seven historically analyzed GSMs were individually checked against source title, disease state, bladder-capacity phenotype, platform and matrix column (four low-capacity IC, three controls); they are the previously seen sensitivity series for the re-audit and are never described as fresh validation.

**GSE621** [5] and **GSE11783** [6] (historical discovery arrays). The audit found two IC-derived mock-APF samples in GSE621 had been mislabeled as normal controls, and GSE11783's ten IC libraries nest in five patients; both corrections are applied before any re-analysis, and the consequences are reported as negatives.

### 3.3 Record roles and nesting

Manifest roles are explicit about dependence. GSE238208's 63 GSM are paired or comparator libraries from 38 people and are used as 25 patient pairs plus a 13-sample block, never as 63 independent observations. GSE11783's ten IC libraries nest in five patients and enter the re-audit as such. GSE621 contributes twelve patient samples and two IC-derived mock-APF samples whose historical "normal control" labels were wrong; those two records are kept with their corrected labels and the misclassification is reported as an audit finding. GSE57560's seven records are the previously seen sensitivity set. No record is counted twice, and no library is promoted to a person.

## 4 The corrected two-source replication is negative

After the audit corrections, discovery was restricted to GSE621 and GSE11783 with mislabeled and nested samples handled, and sensitivity was recomputed in the previously seen GSE57560 (so it cannot be called fresh independent replication). Of 3,326 genes measured in both sources, the new top-50 shares zero genes with the old top-50; the best new-cohort BH value is  $q = .418$  with no gene under .05; and the validation sign agreement is 43/50 (86%) against an empirical null mean of 75.1%,  $p = .0469$ , which fails the registered .025 threshold. The historical positive panel is therefore recorded as a ledger negative, not rewritten into a weaker claim.

### 4.1 Audit and sensitivity arithmetic, exactly as computed

The reproducibility detail a reviewer needs to rerun this verdict is as follows. The frozen historical result (T1) stands on the prior sample-level discovery: 40/40 sign agreements in the original 6-vs-8 and 10-vs-6 contrasts, left intact as a historical artifact. Post-audit, discovery was restricted to the corrected GSE621 and GSE11783 and required evidence from BOTH studies ( $k = 2$ ), leaving 3,326 genes measured in both. On that intersection: the new top-50 by effect shares zero genes with the old top-50; the best new-cohort Benjamini-Hochberg value is  $q = .418$  and no gene falls under .05. The GSE57560 sensitivity then scores the new top-50 in 4 case and 3 control libraries: sign agreement is 43/50 (86%) against an empirical null mean of 75.1% ( $p = .0469$ ), which fails the registered .025 bar, and genes with nominal direction-consistent  $p < .05$  number 5 against a null mean of 4.02 ( $p = .375$ ). Both sensitivity statistics sit inside their nulls. Two caveats are kept with the numbers: GSE57560 was previously seen, so this is a sensitivity, never a fresh replication; and a 4-vs-3 library contrast is too small to support any classifier-utility claim even had it passed. A 2025 publication used overlapping studies, which is why the audit trail, not the literature, is the authority for what the old panel does.

## 5 Preregistered module framework (P48)

The redirect after the negative re-audit was preregistered before any testing. Six candidate biology modules and one known-biology covariate module were locked with fixed gene sets (Table 3) from public libraries (MSigDB Hallmark via Enrichr; GO Biological Process 2025 for the pain module, GO:0019233, 21 genes). The design uses the paired structure of GSE238208: 25 HIC patients each contribute a lesion and a non-lesion biopsy, and the 13 BCG samples form a fixed comparator block.

For training fold  $T$  and gene  $g$ , per-gene  $z$ -scores are computed over the 48 training HIC columns only:

$$z_{g,s} = \frac{x_{g,s} - \bar{x}_{g,T}}{\hat{\sigma}_{g,T}}. \quad (1)$$

Module  $m$ 's per-patient paired effect is the mean member-gene  $z$  difference:

$$\Delta_{m,i} = \bar{z}_{m,\text{lesion},i} - \bar{z}_{m,\text{nonlesion},i}, \quad e_m = \frac{1}{|T|} \sum_{i \in T} \Delta_{m,i}. \quad (2)$$

A module enters the Gate-1 score with weight +1 iff its mean paired effect is positive and a one-sided paired  $t$ -test over training patients gives  $p < .05$ ; the held-out patient score is

$$D_i = \sum_{m: \text{selected}} \Delta_{m,i}. \quad (3)$$

M0 (interferon- $\gamma$  inflammatory) is an excluded covariate: if M0 alone passes and M1–M6 fails, the result is declared known-inflammatory rediscovery, a novelty negative by construction.

## 6 Expression processing and verification protocol

The author FPKM matrix of GSE238208 is used as deposited; no re-quantification was attempted, and the deposited values are treated as the study’s own summary. Within each leave-one-patient-out fold, z-scores are recomputed over the 48 training columns only, so no held-out patient contributes to any normalization constant, module weight, or threshold. Module gene sets are used exactly as committed; coverage is reported per module (Table 3) and never silently filled. For the urinary legs, platform annotation is the bottleneck and is therefore evidenced: A2.3a maps M6 symbols through the full GPL6244 platform table (190/190 recovered), and A2.3b maps the 464 measured MIMAT accessions to mature miRNA names through miRBase mature.fa (2,789 human entries). Every fetch URL is logged in the result JSON at analysis time, per the amendment. ChatGPT consults (rounds 1–2) were used in redirect mode only; verbatim prompts and responses are preserved in `redirect/`, all numerical thresholds were recomputed locally, and one incorrect threshold suggested by the consult ( $\geq 21/25$ ) was corrected to  $\geq 18/25$  with the correction recorded in `redirect/round2_meta.json`.

### 6.1 Inference procedures

All gated statistics were computed with the same three primitives. (1) Leave-one-patient-out scoring: for each of the 25 patients in turn, module gene weights and any reference offsets are refit on the other 24 patients, the held-out patient’s lesion and non-lesion libraries are scored without that patient contributing to any parameter, and  $D$  is the lesion-minus-non-lesion difference of the module composite. (2) Sign inference: the primary statistic is the count of positive  $D$  values, tested by an exact one-sided binomial against  $p_0 = 0.5$ ; the locked pass bar is  $\geq 18/25$  with  $p < 0.025$  (the count tail, not an asymptotic approximation). (3) Effect-size context: a bootstrap 95% confidence interval on the median  $D$  is reported alongside, and for the urine legs - where the design is a group contrast, not a paired sign test - a one-sided permutation null with 20,000 shuffles and fixed seed 20260926. No other inferential machinery is used anywhere in the locked analysis; every number quoted in the gates derives from these primitives applied to the committed fold and composite files.

### 6.2 Module coverage and symbol nomenclature

Of the 200 committed M6 genes, 190 are present in the author FPKM matrix. The ten absent - H2AX, H2AZ1, H2AZ2, H2BC12, JPT1, KMT5A, KNL1, MAP3K20, NSD2, TENT4A - are older or alias symbol forms (for example H2AX vs H2AFX, JPT1 vs HN1, KNL1 vs CASC5) whose nomenclature does not match the matrix’s symbols. No remapping was attempted after the lock: the committed gene list governs, coverage is reported, and the missing ten are treated as unmeasured rather than recovered by post-lock alias judgment. The same rule produced the other modules’ coverage in Table 3.

## 7 Gate 1: the lesion axis is reproducible across held-out patients

Leave-one-patient-out over the 25 HIC patients, refitting every quantity per fold, gives 19/25 held-out patients with  $D_i > 0$  (exact one-sided binomial  $p = .0073$ , threshold .025), median  $D = 0.273$  with bootstrap 95% CI [0.068, 0.644]. The frozen weight rule selected only module M2 (ECM/EMT remodeling) in the training folds. The M0-only novelty control fails (14/25,  $p = .345$ ), so the pass is not known-inflammatory rediscovery. The frozen rule selected M2 in all 25 of 25 training folds (no other module was ever selected), and held-out  $D$  ranges from  $-0.687$  to  $1.546$  with median  $0.273$ . Per-fold values are in Appendix B.

## 8 Gate 2: the reproducible axis is shared with BCG inflammation

Applying the frozen Gate-1 score to the BCG block is negative for Hunner specificity: the score is higher in BCG (0.305) than in Hunner lesions (0.181), Hedges  $g = -0.260$ , permutation  $p = .776$  against the locked pass bar ( $g \geq 0.8$ ,  $p < .025$ ). The replicating lesion axis is therefore severe inflammatory-remodeling biology, not Hunner-specific biology. This is a kept negative, and it motivates the residual design rather than a re-thresholded claim.

## 9 Amendment A2: net-of-BCG residual program

The amendment was locked before any A2 testing. For module  $m$ , the BCG component over the fixed comparator block is

$$b_m = \frac{1}{13} \sum_{s \in \text{BCG}} \bar{z}_{m,s}, \quad (4)$$

and the residual module effect is  $A_m = e_m - b_m$ . A module enters the residual score with weight +1 iff  $A_m > 0$  and the one-sided paired  $t$ -test of  $\Delta_{m,i} - b_m$  over training patients gives  $p < .05$ ; the held-out residual is

$$D_i^{\text{res}} = \sum_{m: \text{selected}} (\Delta_{m,i} - b_m). \quad (5)$$

BCG never enters a HIC training fold; the block is exchangeable.

## 10 Gate A2.1: the G2/M residual is reproducible across held-out patients

### 10.1 Per-patient effect narrative

Across the 25 leave-one-patient-out folds, the Gate 1 lesion-versus-non-lesion contrast (M2) has median  $D = 0.273$  and range  $[-0.687, +1.546]$ , with 19 of 25 patients positive. The strongest individual effects are patients 25 ( $D = 1.546$ ), 18 (1.139) and 13 (1.126); the six negative patients are led in magnitude by 24 ( $-0.687$ ), 9 ( $-0.543$ ) and 21 ( $-0.466$ ). The A2.1 net-of-BCG residual (M6) has median  $D_{\text{resid}} = 0.180$  and range  $[-0.680, +2.047]$ , with 18 of 25 positive. Its three strongest patients are 21 (2.047), 24 (1.158) and 9 (0.888) - precisely the three most negative patients of the M2 contrast. The anti-correspondence is not an artifact of the residual construction: the residual subtracts the fixed BCG offset, identical for every patient, so it cannot invert a per-patient ranking by itself. What the folds show is that the patients whose lesion tissue looks least like inflammatory remodeling are the ones whose tissue shows the strongest G2/M cell-state program relative to BCG inflammation. This is descriptive only; no test was run on the inversion itself.

Only module M6 (G2/M checkpoint, 190 matrix-covered genes) is selected in every fold. The residual contrast meets the locked bar exactly: 18/25 held-out patients positive,  $p = .0216$  ( $P(X \geq 18 \mid 25, .5) = .02164$ ), median residual 0.180, bootstrap 95% CI  $[0.0027, 0.423]$  excluding zero. M6 was the only module selected in any fold (25/25), and held-out residual values range from  $-0.680$  to  $2.047$  with median 0.180. Reading: after removing the module-level biology that BCG inflammatory cystitis also induces, Hunner lesions retain a within-patient, BCG-adjusted, G2/M-associated transcriptional residual that is reproducible across held-out patients in this cohort. This is within-cohort patient-level reproducibility, not independent replication.

## 11 Gate A2.2: specificity bound is compatible, not a validation

The BCG block enters the A2.1 subtraction, so re-testing it cannot be independent; the amendment locks this as a descriptive bound. The residual-adjusted score gives Hedges  $g = 0.454$  (95% CI  $[-0.144, 1.073]$ ),

permutation  $p = .089$ : positive direction, CI overlapping zero - COMPATIBLE, underpowered at 13 BCG samples, and never reported as pass/fail. Stated plainly: the available BCG comparator did not provide evidence sufficient to distinguish the residual from no specificity; the estimate remains compatible with a specificity effect but is not evidence for one, and “bound” here means a very wide interval,  $[-0.144, 1.073]$  in Hedges  $g$  units.

## 12 Patient-level complementarity of the two passes (post-hoc, descriptive)

The two passing gates measure different biology, and their failures fall on different people. Gate 1 is negative for patients 1, 9, 10, 14, 21 and 24; Gate A2.1 is negative for patients 4, 6, 11, 13, 16, 17 and 18; the overlap is empty. Every one of the 25 patients is therefore positive in at least one of the two locked scores, 12 are positive in both, and 13 are positive in exactly one. This is a post-hoc description of committed per-fold values (Appendix B), not a preregistered endpoint, and it must not be quoted as a third gate; what it shows is that the remodeling axis and the G2/M-associated residual are not redundant measurements of one program - the patients without a remodeling signal are exactly the patients carrying the residual signal. A complementary observation in the Gate-1 artifact is that the per-patient M2 and M0 paired effects are essentially uncorrelated (Pearson  $r = -0.078$ ,  $p = 0.712$ ), so the remodeling pass is not shadowing the known interferon- $\gamma$  axis patient-by-patient either.

## 13 Urinary transfer (exploratory)

Two pre-specified exploratory legs test whether the frozen M6 residual program has a noninvasive correlate; per the amendment, no thresholds were tuned on urine data. In urine sediment mRNA (GSE28242; 190/190 M6 symbols mapped via the GPL6244 platform table), Hunner-PBS urine ( $n = 3$ ) scores above lesion-free/controls ( $n = 10$ ): difference 0.286, one-sided permutation  $p = .0406$  - a signal carried substantially by a single high sample (see the per-sample values below), in three donors. In urinary extracellular-vesicle miRNA (GSE196156), the pre-specified validated miRTarBase repressors of the 200-gene M6 module (87,627 human miRNA-target pairs; 262 of the 464 fully measured miRNAs are regulators; the locked direction is lower composite in cases) give composite means  $-0.0273$  (cystitis,  $n = 8$ ) versus  $+0.0525$  (control,  $n = 10$ ), difference  $-0.0798$  in the predicted direction, one-sided permutation  $p = .1144$ . Neither leg meets the .025 bar. A post-hoc descriptive note (`results/ic_p48_a2.urine_descriptive_synthesis.md`) states both directions with exact values and cohort caveats; per the owner ruling it is not a combined test, not a preregistered replication, and no joint  $p$ -value is computed.

### 13.1 Per-sample detail of the two urine legs

In GSE28242 the three with-lesion urines score  $+0.480$ ,  $+0.199$  and  $-0.020$  (mean  $+0.220$ ); the ten lesion-free and control urines range from  $-0.295$  to  $+0.207$  (mean  $-0.066$ ), and two of the five control subjects score above  $+0.12$ . The group difference of 0.286 is therefore carried substantially by one high with-lesion profile (GSM699147) on a three-donor arm; the one-sided permutation  $p = .0406$  is the weakest link in the chain and is reported, not rounded up. In GSE196156 the regulator composite is below the control mean ( $+0.053$ ) in six of eight cystitis profiles (range  $-0.184$  to  $+0.141$ ), and four of ten controls also fall below their own group mean - the direction is present but the groups overlap widely, which is exactly what  $p = .1144$  says. The two BPS profiles average  $-0.153$ , descriptively at least as far in the predicted case direction as the cystitis arm itself; with  $n = 2$  this is anecdote, but it cautions against reading the cystitis-versus-control trend as lesion-specific rather than pain-syndrome-wide.

## 14 External annotation of the residual program

Forty-three external scientific services were genuinely used on this lane (41 under the most conservative re-count of the NCBI e-utils rows), each with repo-resident evidence (Appendix E). Conclusions drawn from them are contextual, not independent replication, because annotation records share underlying evidence bases. The five M6 genes with the most validated miRNA regulation (G3BP1, SRSF1, DR1, RASAL2, MAPK14) are regulatory hubs rather than a physical complex: STRING returns zero physical edges among them. Their druggability is uneven: DGIdb reports 105 interactions for MAPK14 (including p38 $\alpha$  inhibitors) and none for the other hubs, and Pharos classifies MAPK14 as Tchem while the others are Tbio. In TCGA bladder tumors, mutation records are sparse (RASAL2 16, SRSF1 6, MAPK14 4, G3BP1 2, DR1 1). PubMed abstract co-mention of any hub with “interstitial cystitis” is zero, and Open Targets places none of the hubs among the top-50 targets associated with interstitial cystitis (EFO\_1000869): we did not identify prior reports directly testing this locked residual program - a reproducible statement about our searches on this date, not evidence of biological novelty, since annotation coverage is incomplete.

## 15 Gene-level anatomy of the M6 regulator set

### 15.1 Platform coverage of the measured regulator slice

The A2.3b regulator universe is large and densely connected: 87,627 validated human miRNA-target pairs name 2,807 distinct regulators against the 200 M6 genes, and per-gene files show a median of 676.5 distinct regulator families per M6 gene. The families targeting at least 100 of the 200 genes are miR-548 (180 genes), miR-29 (172), miR-15 (166), miR-30 (164), miR-16 (162), miR-103 (153), miR-92 (150), miR-17 (150), miR-24 (147) and miR-181 (144) - broadly pleiotropic families, not M6-specific ones. Against that universe, the urine array measures 262 mature forms. These map to 221 distinct miRNA families; 182 of those families contribute a single measured member, and the largest measured family (miR-4433) contributes four. By arm, 162 measured forms are 5p and 100 are 3p. The measured slice is therefore thin and family-sparse relative to the regulator space the module presents, and the A2.3b negative is a statement about that platform’s slice - not about urinary miRNA regulation of the program in general. A panel with denser family coverage would be a different, separately preregisterable test.

The 87,627 validated human miRNA-target pairs behind A2.3b distribute very unevenly across the 200 M6 genes (per-gene counts committed as `results/ic_p48_a2_m6_gene_mirna_pairs.csv`; 87,632 row-level records, of which 87,627 carry hsa- prefixed miRNA names and entered the regulator set). Median coverage is 400.5 pairs per gene. The five most-targeted genes - G3BP1 (1,289), SRSF1 (1,274), DR1 (1,250), RASAL2 (1,220), MAPK14 (1,165) - are regulatory and signaling hubs, not the canonical mitotic machinery: the core G2/M kinases carry markedly less miRNA annotation (CDK1 487, BIRC5 529, AURKA 325, TTK 223, AURKB 203, PLK1 119, CCNB2 59, CDC20 35). Two readings follow. First, the composite used in A2.3b is dominated by heavily annotated hub genes, so its direction reflects regulator coverage of hub biology more than of the mitotic core - one more reason the urine legs stay exploratory. Second, the annotation gradient is itself evidence about where miRNA regulation of the G2/M program has been studied: stress-granule and splicing regulators (G3BP1, SRSF1) and the p38 kinase MAPK14 are far better miRNA-annotated than the checkpoint kinases, which matches their broader multi-pathway roles and explains why DGIdb and Pharos flag only MAPK14 as chemically tractable (Tchem) among the hubs.

## 16 Published reference points

The published expression literature for Hunner-lesion IC is small, which is why this lane treats every comparison as landscape rather than benchmark. The primary reference point is the source study of GSE238208 [1], which established the interferon- $\gamma$ /Th1-17 polarization of HIC tissue and motivated the M0 covariate design here: our framework asks what survives after exactly that known axis is excluded, and the M0 nov-

elty control confirms the known axis alone does not reproduce the Gate-1 pass. The older array literature [5, 6, 4] is the basis of the historical panel whose corrected re-audit is reported as negative above. The urine-sediment study behind GSE28242 [3] evaluated noninvasive mRNA markers and is the direct precedent for the A2.3a transfer leg; its with-lesion arm is the same three-donor group that bounds our exploratory signal. No published locked-protocol comparator exists for the residual-program question asked here, so no same-task benchmark win is claimed or implied; a formal published comparator test remains an open gate of the project.

## 17 Published comparator: the Zhou 2025 marker panel (P50)

The NOVELTY\_PLAN requires a frozen evaluation against a named published baseline before any same-task advantage is claimed. The baseline is Zhou et al.’s 2025 IC/BPS marker panel [16]: PLAC8, S100A8 and PPBP, all up-regulated in IC/BPS tissue in their study (per-gene AUCs 0.887, 0.818, 0.871 on pooled GSE11783, GSE28242 and GSE57560 data). That study did not use GSE238208, so this cohort is untouched by the baseline. Because the publication reports genes and directions but no weights, the frozen evaluable form is an unweighted mean of per-gene  $z$ -scores, scored with exactly the LOPO machinery of Gates 1 and A2.1; the protocol, endpoints and the win rule were locked in `prereg/IC.P50_published_comparator.md` before the first (and only) run.

The published panel reproduces on the untouched cohort: 19 of 25 held-out patients positive ( $p = .0073$ , median  $D = 0.184$ ). Contrary to the registered prediction, the panel is not BCG-shared in this matched construction: its BCG offset is  $-0.680$  (BCG samples score below the HIC reference on these genes), and the BCG-adjusted discrimination is stronger, not weaker (24/25,  $p = 7.7 \times 10^{-7}$ ). Correlations across patients are  $r = -0.29$  ( $p = .157$ ) against the M2 Gate 1 values and  $r = +0.338$  ( $p = .098$ ) against the M6 residual.

The frozen win rule therefore fails on two of its three legs - the adjusted panel is stronger than the raw panel, and the panel-residual correlation exceeds the pre-locked 0.3 bound - so no same-task advantage is claimed. What the comparison establishes instead: the Zhou panel is real and generalizes to an independent cohort; it is not redundant with the remodeling axis; and its mild positive correlation with the residual means the two signals overlap more than the locked orthogonality bar allows. The residual program’s contribution is the BCG-adjustment framing itself, not superiority over the published marker. The registered prediction that the inflammatory panel members would collapse after BCG adjustment was wrong and is retained in the ledger as a wrong prediction, not edited away.

## 18 Discussion

The biological reading across the locked gates is consistent. Hunner lesions carry a strong remodeling/inflammatory axis (M2) that replicates across patients but is equally a BCG-cystitis program; treating it as a Hunner marker would mislabel severe inflammation generally, and the locked Gate-2 negative prevents exactly that error. Removing the BCG-shared component leaves a BCG-adjusted, G2/M-associated transcriptional residual that is reproducible across held-out patients within this cohort (A2.1). Whether it reflects increased proliferation, a shifted epithelial/immune/stromal cell mixture, or another cell state cannot be resolved from bulk RNA; the hyperproliferation reading is a hypothesis whose direct tests are single-cell or spatial transcriptomics and proliferation-marker immunostaining, not more bulk scoring. The two urinary legs are inconclusive rather than confirmatory. The honest summary is: a within-cohort reproducible tissue residual program; the urinary path is unproven.

### 18.1 What the M0 control does and does not rule out

The novelty control asks whether a known IFN $\gamma$ -inflammatory module (M0) alone would have passed the Gate 1 bar. It does not: 14 of 25 patients positive ( $p = 0.345$ ), median  $D = 0.012$ . So the M2 pass (19/25)

is not explained by the generic observation that inflamed tissue looks inflamed - the canonical inflammatory program does not itself separate lesion from non-lesion in this cohort. The post-hoc per-patient correlation between the M2 and M0 effects is  $r = -0.078$  ( $p = 0.71$ ), meaning the patients who drive the M2 pass are not the patients with the strongest inflammatory-module contrast. What the M0 control cannot rule out is the reverse direction: non-lesion tissue in a Hunner bladder may itself be inflamed, compressing any inflammation-based contrast. The control's failure is therefore evidence about module identity, not about the biology of the non-lesion state.

## 18.2 What an independent validation cohort requires

The falsifiable contribution of this work is a claim about what a future cohort must show, and the cohort requirements follow directly from how the current gates are built. First, paired Hunner-lesion and non-lesion tissue from the same bladders, with patient as the unit of inference - the LOPO design exists because libraries are not independent of donors. Second, BCG-exposure metadata recorded per patient, since the residual program is defined against a BCG-inflammation offset and a cohort without exposure data cannot replicate the net-of-BCG construction. Third, the module membership, weights, offset and endpoint locked in a preregistration before the cohort is unblinded or, for a public accession, before its matrix is opened - the A2.3b episode in this project is a worked example of why: two individually exploratory directional results do not license a post-hoc combination, and any combined-urine variant requires an untouched cohort with fixed weights and a fixed endpoint locked in advance. Fourth, coverage: at least 190 of the 200 M6 genes measurable, and a regulator panel denser than 262 forms if the urinary transfer is to be retested. A cohort meeting those four requirements turns the residual program from a replicated-in-one-cohort finding into a testable prediction.

## 18.3 Where this sits in the published landscape

Three published anchors frame the claims. The AUA/SUFU guideline keeps IC/BPS a clinical diagnosis made on symptoms and cystoscopy findings, not on a molecular test [12]. The systematic review of the Hunner phenotype supports treating HIC as a distinct entity from non-lesion BPS [11]. And the 2021 biomarker review finds no validated molecular biomarker for either phenotype and catalogs candidate markers that remain research-grade [13]. Against that landscape, nothing in this manuscript is a validated biomarker: the residual program is a within-cohort reproducible molecular correlate, and the urinary legs are research assays at most. The source studies behind the accessions used here [1, 5, 6, 3, 4, 2] are the evidentiary base; the reviews are context, not support.

## 18.4 Module-level reading of the two gates

The two positive gates select different modules for consistent reasons. Gate 1 rewards the largest reproducible lesion-versus-non-lesion contrast, which is the ECM/EMT remodeling module M2: fibrotic remodeling is the dominant within-patient axis in HIC tissue, but the BCG block shows the same axis, so it carries no Hunner specificity. The residual design subtracts exactly this shared component at module level; what remains is M6, a G2/M-associated transcriptional program. That the residual is G2/M-associated rather than barrier-specific (M1), metabolic (M4), stress-related (M5) or pain-related (M3) is itself an empirical result of the locked selection rule, not a modeling choice: in 25/25 folds no other module cleared the residual bar. The pain module M3 (21 genes, GO:0019233) never enters either score, consistent with lesion biology being separable from pain biology in this cohort. The interferon- $\gamma$  module M0 fails its novelty control, which keeps the positive gates from being reinterpreted as rediscovery of the known Th1/17 axis [1].

## 19 Limitations

### 19.1 Cell composition is an untested alternative explanation

Every score in this manuscript is bulk-tissue or bulk-fluid RNA. A lesion-versus-non-lesion biopsy contrast is also a spatial sampling contrast: Hunner lesions are histologically abnormal regions, so a module difference can reflect a changed epithelial fraction, immune infiltrate, stromal abundance, or lesion architecture rather than a changed transcriptional program within cells. Both positive gates are exposed to this: the M2 remodeling axis could be more stroma, and the M6 G2/M-associated residual could be more cycling cells of any type rather than a urothelial state. No deconvolution was performed (the cohort ships expression matrices without matched cell counts), so cell composition stands as an untested alternative explanation, and the single-cell/spatial and immunostaining experiments named in the Discussion are the direct adjudicators.

### 19.2 Cohort, platform and annotation bounds

Each dataset bounds what can be said. GSE238208 is 25 patients from one study and one platform; the residual replication is therefore a within-study patient-level result, and between-study heterogeneity is unestimated. The BCG comparator is 13 samples, which is why the A2.2 specificity bound can only be “compatible” rather than informative. The GSE28242 with-lesion arm is three donors, so A2.3a’s directional trend has a wide null. On the EV-miRNA side, isolation and quantification choices materially move urinary EV transcriptomic readouts [14], which bounds how much weight any one array’s composite can carry. The GSE196156 cystitis arm is eight profiles, the composite averages 262 measured regulators - a broad set that dilutes any per-miRNA signal - and 2,168 deposited rows were detection-gated out by the locked measured-set definition. The A2.2 bound is not independent by construction. Per-gene external annotations are correlated through shared evidence bases (UniProt, GO, GEO), so they cannot be summed as independent support. The historical sensitivity reuses the previously seen GSE57560, so it cannot be described as fresh replication. The cultured-cell series GSE11839 is context only. Nothing here is a clinical test, the exploratory urine legs must not be quoted without their threshold failures, the descriptive urine synthesis must not be quoted as a combined test, and the two BPS urine-EV profiles trend in the same direction as the cystitis arm, which any future lesion-specificity claim would have to confront.

Table 1: Individually verified sample records by series and group.

| Accession  | Series    | Title                                                  | Group                    |
|------------|-----------|--------------------------------------------------------|--------------------------|
| GSM298203  | GSE11783  | patient:10–location:healthy–R:01                       |                          |
| GSM298204  | GSE11783  | patient:11–location:healthy–R:01                       |                          |
| GSM298205  | GSE11783  | patient:12–location:ni–R:01                            |                          |
| GSM298206  | GSE11783  | patient:12–location:ulcus–R:01                         |                          |
| GSM298207  | GSE11783  | patient:13–location:ni–R:01                            |                          |
| GSM298208  | GSE11783  | patient:13–location:ulcus–R:01                         |                          |
| GSM298209  | GSE11783  | patient:14–location:ulcus–R:01                         |                          |
| GSM298210  | GSE11783  | patient:15–location:ni–R:01                            |                          |
| GSM298211  | GSE11783  | patient:15–location:ulcus–R:01                         |                          |
| GSM298212  | GSE11783  | patient:2–location:healthy–R:01                        |                          |
| GSM298213  | GSE11783  | patient:4–location:ni–R:01                             |                          |
| GSM298214  | GSE11783  | patient:4–location:ulcus–R:01                          |                          |
| GSM298215  | GSE11783  | patient:5–location:healthy–R:01                        |                          |
| GSM298216  | GSE11783  | patient:7–location:healthy–R:01                        |                          |
| GSM298217  | GSE11783  | patient:9–location:healthy–R:01                        |                          |
| GSM402541  | GSE11783  | patient:14–location:ni–R:01                            |                          |
| GSM299095  | GSE11839  | human cultured bladder HB DM, subject 8                | control cultured urothel |
| GSM299096  | GSE11839  | human cultured bladder HB DM, subject 12               | control cultured urothel |
| GSM299097  | GSE11839  | human cultured bladder HB DM, subject 13               | control cultured urothel |
| GSM299098  | GSE11839  | human cultured bladder HB KM, subject 8                | control cultured urothel |
| GSM299099  | GSE11839  | human cultured bladder HB KM, subject 12               | control cultured urothel |
| GSM299100  | GSE11839  | human cultured bladder HB KM, subject 13               | control cultured urothel |
| GSM299101  | GSE11839  | human cultured bladder IC DM, subject 1                | IC cultured urothelial   |
| GSM299102  | GSE11839  | human cultured bladder IC DM, subject 3                | IC cultured urothelial   |
| GSM299103  | GSE11839  | human cultured bladder IC DM, subject 10               | IC cultured urothelial   |
| GSM299104  | GSE11839  | human cultured bladder IC KM, subject 1                | IC cultured urothelial   |
| GSM299105  | GSE11839  | human cultured bladder IC KM, subject 3                | IC cultured urothelial   |
| GSM299106  | GSE11839  | human cultured bladder IC KM, subject 10               | IC cultured urothelial   |
| GSM5861368 | GSE196156 | Urine_Cystitis_1                                       | Cystitis                 |
| GSM5861369 | GSE196156 | Urine_Cystitis_2                                       | Cystitis                 |
| GSM5861370 | GSE196156 | Urine_Cystitis_3                                       | Cystitis                 |
| GSM5861371 | GSE196156 | Urine_Cystitis_4                                       | Cystitis                 |
| GSM5861372 | GSE196156 | Urine_Cystitis_5                                       | Cystitis                 |
| GSM5861373 | GSE196156 | Urine_Cystitis_6                                       | Cystitis                 |
| GSM5861374 | GSE196156 | Urine_Cystitis_7                                       | Cystitis                 |
| GSM5861375 | GSE196156 | Urine_Cystitis_8                                       | Cystitis                 |
| GSM5861376 | GSE196156 | Urine_BPS_1                                            | BPS                      |
| GSM5861377 | GSE196156 | Urine_BPS_2                                            | BPS                      |
| GSM5861378 | GSE196156 | Urine_Control_1                                        | Control                  |
| GSM5861379 | GSE196156 | Urine_Control_2                                        | Control                  |
| GSM5861380 | GSE196156 | Urine_Control_3                                        | Control                  |
| GSM5861381 | GSE196156 | Urine_Control_4                                        | Control                  |
| GSM5861382 | GSE196156 | Urine_Control_5                                        | Control                  |
| GSM5861383 | GSE196156 | Urine_Control_6                                        | Control                  |
| GSM5861384 | GSE196156 | Urine_Control_7                                        | Control                  |
| GSM5861385 | GSE196156 | Urine_Control_8                                        | Control                  |
| GSM5861386 | GSE196156 | Urine_Control_9                                        | Control                  |
| GSM5861387 | GSE196156 | Urine_Control_10                                       | Control                  |
| GSM7660353 | GSE238208 | Hunner_Type_Interstitial_Cystitis_Case1_Hunner_Lesion  | Hunner lesion            |
| GSM7660354 | GSE238208 | Hunner_Type_Interstitial_Cystitis_Case2_Hunner_Lesion  | Hunner lesion            |
| GSM7660355 | GSE238208 | Hunner_Type_Interstitial_Cystitis_Case3_Hunner_Lesion  | Hunner lesion            |
| GSM7660356 | GSE238208 | Hunner_Type_Interstitial_Cystitis_Case4_Hunner_Lesion  | Hunner lesion            |
| GSM7660357 | GSE238208 | Hunner_Type_Interstitial_Cystitis_Case5_Hunner_Lesion  | Hunner lesion            |
| GSM7660358 | GSE238208 | Hunner_Type_Interstitial_Cystitis_Case6_Hunner_Lesion  | Hunner lesion            |
| GSM7660359 | GSE238208 | Hunner_Type_Interstitial_Cystitis_Case7_Hunner_Lesion  | Hunner lesion            |
| GSM7660360 | GSE238208 | Hunner_Type_Interstitial_Cystitis_Case8_Hunner_Lesion  | Hunner lesion            |
| GSM7660361 | GSE238208 | Hunner_Type_Interstitial_Cystitis_Case9_Hunner_Lesion  | Hunner lesion            |
| GSM7660362 | GSE238208 | Hunner_Type_Interstitial_Cystitis_Case10_Hunner_Lesion | Hunner lesion            |
| GSM7660363 | GSE238208 | Hunner_Type_Interstitial_Cystitis_Case11_Hunner_Lesion | Hunner lesion            |
| GSM7660364 | GSE238208 | Hunner_Type_Interstitial_Cystitis_Case12_Hunner_Lesion | Hunner lesion            |
| GSM7660365 | GSE238208 | Hunner_Type_Interstitial_Cystitis_Case13_Hunner_Lesion | Hunner lesion            |

Table 3: Locked modules (gene counts; matrix coverage in parentheses).

| Module                     | Source term                             | <i>n</i> genes      | Role                                                         |
|----------------------------|-----------------------------------------|---------------------|--------------------------------------------------------------|
| M0_known_inflammatory      | Interferon Gamma Response               | 200                 | known-biology covariate; EXCLUDED from the novelty score (EX |
| M1_barrier_apical_junction | Apical Junction                         | 200 (193 in matrix) | candidate novelty module: epithelial barrier                 |
| M2_ecm_remodeling          | Epithelial Mesenchymal Transition       | 200 (197 in matrix) | candidate novelty module: ECM/remodeling                     |
| M3_neuronal_sensory_pair   | Sensory Perception of Pain (GO:0019233) | 21 (20 in matrix)   | candidate novelty module: neuronal/sensory                   |
| M4_metabolism_oxphos       | Oxidative Phosphorylation               | 200 (184 in matrix) | candidate novelty module: metabolism                         |
| M5_stress_repair_hypoxia   | Hypoxia                                 | 200 (192 in matrix) | candidate novelty module: stress/repair                      |
| M6_cell_state_g2m          | G2-M Checkpoint                         | 200 (190 in matrix) | candidate novelty module: cell-state/proliferation           |

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## A Derivations

The binomial gate uses the exact one-sided tail. For  $n$  independent patients with null success probability  $1/2$ ,

$$P(X \geq k | n) = \sum_{j=k}^n \binom{n}{j} 2^{-n}, \quad (6)$$

so  $P(X \geq 19 | 25) = 0.00732$  and  $P(X \geq 18 | 25) = 0.02164$ ; the locked bar is .025 one-sided, matching the program's two-sided .05 convention. Hedges'  $g$  corrects the standardized mean difference for small samples:

$$d = \frac{\bar{x}_1 - \bar{x}_0}{s_p}, \quad g = Jd, \quad J = 1 - \frac{3}{4(n_1 + n_0) - 9}. \quad (7)$$

The permutation  $p$  for a one-sided mean difference  $\Delta$  over  $B$  label reshuffles is the  $(+1)$ -smoothed tail fraction

$$p = \frac{1 + \#\{b : \Delta_b \text{ at least as extreme as } \Delta_{\text{obs}}\}}{1 + B}, \quad (8)$$

which is never zero and is exact for the exchangeable null. The residual construction (Eq. 5–6) is a module-space analogue of partialling out a comparator-induced component: because  $b_m$  is a constant within a fold, the paired  $t$ -test of  $\Delta_{m,i} - b_m$  is the paired test of  $\Delta_{m,i}$  against the shifted null  $b_m$ , so BCG contributes a fixed offset, never variance, to the patient-level inference.

### A.1 A fully worked fold: patient 25 under Gate 1

One complete fold, end to end, from the committed values (`results/ic_p48_gate1_fold_detail.csv`). Patient 25 is held out; module weights and reference distributions are refit on the other 24 patients with the locked procedure, and the held-out pair is scored with no parameter seeing patient 25. The M2 composite scores are  $S_{\text{lesion}} = +0.9777$  and  $S_{\text{nonlesion}} = -0.5683$ , giving  $D = +1.5460$  - the largest positive fold in the gate (range  $[-0.6874, +1.5460]$ ). The fold contributes one to the count of positives; the count reaches 19, and the exact one-sided binomial tail  $P(X \geq 19 | n = 25, p = 0.5) = 0.00732$  is the Gate 1  $p$ -value. Under A2.1 the same held-out machinery on the BCG-adjusted residual gives patient 21 the largest value,  $D_{\text{resid}} = +2.0467$  (median 0.1798), and the count of positives is 18 with tail  $P(X \geq 18 | 25, .5) = 0.02164$ . No other statistic enters either gate.

### A.2 Worked tail computations

With  $n = 25$  paired patients and null per-patient success probability  $1/2$ , the exact one-sided Gate-1 tail is

$$P(X \geq 19 | 25) = 2^{-25} \sum_{j=19}^{25} \binom{25}{j} = \frac{184,756 + 53,130 + 23,000 + 5,200 + 2,300 + 126 + 25 + 1 \text{ terms}}{33,554,432} = 0.00732. \quad (9)$$

The A2.1 tail is  $P(X \geq 18 | 25) = 0.02164$ ; both are below the locked .025 bar, while  $P(X \geq 17 | 25) = 0.0539$  would not have been - so the locked threshold  $k = 18$  is the smallest passing count, and the M0-only control ( $14/25, p = 0.345$ ) fails by a wide margin. These integers, not asymptotics, are the test.

### A.3 Why the permutation null is the right null here

The LOPO sign test assumes only that, under no lesion program, the sign of each held-out  $D_i$  is an independent fair coin; pairing removes all between-patient expression level differences by construction. For the two-sample permutation tests (Gate 2, A2.2, both urine legs), the null is exchangeability of group labels; the BCG block is a fixed comparator in the amendment design, so in A2.2 only the spread of the bound is informative, and the amendment locks that reading before computation.

#### A.4 The residual as a fixed offset

Because  $b_m$  is constant within a fold, testing  $\Delta_{m,i} - b_m > 0$  over patients is the location-shifted paired test of  $\Delta_{m,i}$  against  $b_m$ :

$$\text{Var}_i(\Delta_{m,i} - b_m) = \text{Var}_i(\Delta_{m,i}), \quad t = \frac{\overline{\Delta_m} - b_m}{s_{\Delta_m} / \sqrt{n_T}}, \quad (10)$$

so the BCG block contributes a fixed offset to the numerator and nothing to the variance. This is the module-space analogue of partialling out a comparator-induced component, and it is why A2.2 cannot recycle the same block as an independent validation set.

## B Per-fold held-out values

### Gate 1 folds

| held_out_patient | modules_used    | S_lesion | S_nonlesion | D       |
|------------------|-----------------|----------|-------------|---------|
| 1                | M2_ecm_modeling | -0.4105  | -0.2845     | -0.126  |
| 2                | M2_ecm_modeling | 0.895    | -0.1606     | 1.0555  |
| 3                | M2_ecm_modeling | -0.3774  | -0.8103     | 0.4329  |
| 4                | M2_ecm_modeling | 0.3802   | -0.6533     | 1.0335  |
| 5                | M2_ecm_modeling | 0.4063   | 0.187       | 0.2193  |
| 6                | M2_ecm_modeling | 0.4532   | -0.1544     | 0.6076  |
| 7                | M2_ecm_modeling | 0.6771   | 0.4041      | 0.273   |
| 8                | M2_ecm_modeling | -0.0886  | -0.1571     | 0.0685  |
| 9                | M2_ecm_modeling | -0.5234  | 0.02        | -0.5434 |
| 10               | M2_ecm_modeling | 0.2467   | 0.6331      | -0.3864 |
| 11               | M2_ecm_modeling | -0.0354  | -0.7767     | 0.7414  |
| 12               | M2_ecm_modeling | 0.3245   | 0.2752      | 0.0494  |
| 13               | M2_ecm_modeling | 0.4563   | -0.6693     | 1.1256  |
| 14               | M2_ecm_modeling | -0.8056  | -0.7532     | -0.0524 |
| 15               | M2_ecm_modeling | -0.1797  | -0.3765     | 0.1968  |
| 16               | M2_ecm_modeling | 0.6742   | 0.0303      | 0.6439  |
| 17               | M2_ecm_modeling | -0.7431  | -0.9393     | 0.1961  |
| 18               | M2_ecm_modeling | 0.8501   | -0.2891     | 1.1392  |
| 19               | M2_ecm_modeling | 0.8193   | 0.2041      | 0.6152  |
| 20               | M2_ecm_modeling | 1.0216   | -0.0016     | 1.0232  |
| 21               | M2_ecm_modeling | -0.1576  | 0.3086      | -0.4662 |
| 22               | M2_ecm_modeling | 0.3673   | -0.089      | 0.4563  |
| 23               | M2_ecm_modeling | 0.1446   | 0.0391      | 0.1055  |
| 24               | M2_ecm_modeling | -0.6606  | 0.0268      | -0.6874 |
| 25               | M2_ecm_modeling | 0.9777   | -0.5683     | 1.546   |

### Gate A2.1 folds

| held_out_patient | modules_used      | D_residual |
|------------------|-------------------|------------|
| 1                | M6_cell_state_g2m | 0.0027     |
| 2                | M6_cell_state_g2m | 0.5956     |
| 3                | M6_cell_state_g2m | 0.7106     |
| 4                | M6_cell_state_g2m | -0.6798    |
| 5                | M6_cell_state_g2m | 0.1401     |

| held_out_patient | modules_used      | D_residual |
|------------------|-------------------|------------|
| 6                | M6_cell_state_g2m | -0.0815    |
| 7                | M6_cell_state_g2m | 0.129      |
| 8                | M6_cell_state_g2m | 0.1798     |
| 9                | M6_cell_state_g2m | 0.8881     |
| 10               | M6_cell_state_g2m | 0.501      |
| 11               | M6_cell_state_g2m | -0.5039    |
| 12               | M6_cell_state_g2m | 0.471      |
| 13               | M6_cell_state_g2m | -0.0189    |
| 14               | M6_cell_state_g2m | 0.4234     |
| 15               | M6_cell_state_g2m | 0.1956     |
| 16               | M6_cell_state_g2m | -0.0029    |
| 17               | M6_cell_state_g2m | -0.2276    |
| 18               | M6_cell_state_g2m | -0.6137    |
| 19               | M6_cell_state_g2m | 0.2694     |
| 20               | M6_cell_state_g2m | 0.1531     |
| 21               | M6_cell_state_g2m | 2.0467     |
| 22               | M6_cell_state_g2m | 0.2421     |
| 23               | M6_cell_state_g2m | 0.146      |
| 24               | M6_cell_state_g2m | 1.1582     |
| 25               | M6_cell_state_g2m | 0.29       |

## C Locked module gene sets

Gene lists as committed in `prereg/ic_p48_module_genesets.json`.

### M0.known\_inflammatory\_IFNg (200 genes, Interferon Gamma Response)

|          |         |         |          |
|----------|---------|---------|----------|
| ADAR     | APOL6   | ARID5B  | ARL4A    |
| AUTS2    | B2M     | BANK1   | BATF2    |
| BPGM     | BST2    | BTG1    | C1R      |
| C1S      | CASP1   | CASP3   | CASP4    |
| CASP7    | CASP8   | CCL2    | CCL5     |
| CCL7     | CD274   | CD38    | CD40     |
| CD69     | CD74    | CD86    | CDKN1A   |
| CFB      | CFH     | CIITA   | CMKLR1   |
| CMPK2    | CMTR1   | CSF2RB  | CXCL10   |
| CXCL11   | CXCL9   | DDX58   | DDX60    |
| DHX58    | EIF2AK2 | EIF4E3  | EPSTI1   |
| FAS      | FCGR1A  | FGL2    | FPR1     |
| GBP4     | GBP6    | GCH1    | GPR18    |
| GZMA     | HELZ2   | HERC6   | HIF1A    |
| HLA-A    | HLA-B   | HLA-DMA | HLA-DQA1 |
| HLA-DRB1 | HLA-G   | ICAM1   | IDO1     |
| IFI27    | IFI30   | IFI35   | IFI44    |
| IFI44L   | IFIH1   | IFIT1   | IFIT2    |
| IFIT3    | IFITM2  | IFITM3  | IFNAR2   |
| IL10RA   | IL15    | IL15RA  | IL18BP   |
| IL2RB    | IL4R    | IL6     | IL7      |
| IRF1     | IRF2    | IRF4    | IRF5     |

|          |          |         |          |
|----------|----------|---------|----------|
| IRF7     | IRF8     | IRF9    | ISG15    |
| ISG20    | ISOC1    | ITGB7   | JAK2     |
| KLRK1    | LAP3     | LATS2   | LCP2     |
| LGALS3BP | LY6E     | LYSMD2  | MARCHF1  |
| METTL7B  | MT2A     | MTHFD2  | MVP      |
| MX1      | MX2      | MYD88   | NAMPT    |
| NCOA3    | NFKB1    | NFKBIA  | NLRC5    |
| NMI      | NOD1     | NUP93   | OAS2     |
| OAS3     | OASL     | OGFR    | P2RY14   |
| PARP12   | PARP14   | PDE4B   | PELI1    |
| PFKP     | PIM1     | PLA2G4A | PLSCR1   |
| PML      | PNP      | PNPT1   | PSMA2    |
| PSMA3    | PSMB10   | PSMB2   | PSMB8    |
| PSMB9    | PSME1    | PSME2   | PTGS2    |
| PTPN1    | PTPN2    | PTPN6   | RAPGEF6  |
| RBCK1    | RIPK1    | RIPK2   | RNF213   |
| RNF31    | RSAD2    | RTP4    | SAMD9L   |
| SAMHD1   | SECTM1   | SELP    | SERPING1 |
| SLAMF7   | SLC25A28 | SOCS1   | SOCS3    |
| SOD2     | SP110    | SPPL2A  | SRI      |
| SSPN     | ST3GAL5  | ST8SIA4 | STAT1    |
| STAT2    | STAT3    | STAT4   | TAP1     |
| TAPBP    | TDRD7    | TNFAIP2 | TNFAIP3  |
| TNFAIP6  | TNFSF10  | TOR1B   | TRAFD1   |
| TRIM14   | TRIM21   | TRIM25  | TRIM26   |
| TXNIP    | UBE2L6   | UPP1    | USP18    |
| VAMP5    | VAMP8    | VCAM1   | WARS1    |
| XAF1     | XCL1     | ZBP1    | ZNFX1    |

#### **M1\_barrier\_apical\_junction (200 genes, Apical Junction)**

|         |         |        |         |
|---------|---------|--------|---------|
| ACTA1   | ACTB    | ACTC1  | ACTG1   |
| ACTG2   | ACTN1   | ACTN2  | ACTN3   |
| ACTN4   | ADAM15  | ADAM23 | ADAM9   |
| ADAMTS5 | ADRA1B  | AKT2   | AKT3    |
| ALOX15B | AMH     | AMIGO1 | AMIGO2  |
| ARHGEF6 | ARPC2   | ATP1A3 | B4GALT1 |
| BAIAP2  | BMP1    | CADM2  | CADM3   |
| CALB2   | CAP1    | CD209  | CD274   |
| CD276   | CD34    | CD86   | CD99    |
| CDH1    | CDH11   | CDH15  | CDH3    |
| CDH4    | CDH6    | CDH8   | CDK8    |
| CDSN    | CERCAM  | CLDN11 | CLDN14  |
| CLDN15  | CLDN18  | CLDN19 | CLDN4   |
| CLDN5   | CLDN6   | CLDN7  | CLDN8   |
| CLDN9   | CNN2    | CNTN1  | COL16A1 |
| COL17A1 | COL9A1  | CRAT   | CRB3    |
| CTNNA1  | CTNND1  | CX3CL1 | DHX16   |
| DLG1    | DMP1    | DSC1   | DSC3    |
| EGFR    | EPB41L2 | EVL    | EXOC4   |
| FBN1    | FLNC    | FSCN1  | FYB1    |

|           |         |         |         |
|-----------|---------|---------|---------|
| GAMT      | GNAI1   | GNAI2   | GRB7    |
| GTF2F1    | HADH    | HRAS    | ICAM1   |
| ICAM2     | ICAM4   | ICAM5   | IKBKG   |
| INPPL1    | INSIG1  | IRS1    | ITGA10  |
| ITGA2     | ITGA3   | ITGA9   | ITGB1   |
| ITGB4     | JAM3    | JUP     | KCNH2   |
| KRT31     | LAMA3   | LAMB3   | LAMC2   |
| LAYN      | LDLRAP1 | LIMA1   | MADCAM1 |
| MAP3K20   | MAP4K2  | MAPK11  | MAPK13  |
| MAPK14    | MDK     | MMP2    | MMP9    |
| MPP5      | MPZL1   | MPZL2   | MSN     |
| MVD       | MYH10   | MYH9    | MYL12B  |
| MYL9      | NECTIN1 | NECTIN2 | NECTIN3 |
| NECTIN4   | NEGR1   | NEXN    | NF1     |
| NF2       | NFASC   | NLGN2   | NLGN3   |
| NRAP      | NRTN    | NRXN2   | PARD6G  |
| PARVA     | PBX2    | PCDH1   | PDZD3   |
| PECAM1    | PFN1    | PIK3CB  | PIK3R3  |
| PKD1      | PLCG1   | PPP2R2C | PTEN    |
| PTK2      | PTPRC   | RAC2    | RASA1   |
| RHOF      | RRAS    | RSU1    | SDC3    |
| SGCE      | SHC1    | SHROOM2 | SIRPA   |
| SKAP2     | SLC30A3 | SLIT2   | SORBS3  |
| SPEG      | SRC     | STX4    | SYK     |
| SYMPK     | TAOK2   | TGFBI   | THBS3   |
| THY1      | TIAL1   | TJP1    | TMEM8B  |
| TNFRSF11B | TRAF1   | TRO     | TSC1    |
| TSPAN4    | TUBG1   | VASP    | VAV2    |
| VCAM1     | VCAN    | VCL     | VWF     |
| WASL      | WNK4    | YWHAH   | ZYX     |

## M2\_ecm\_remodeling (200 genes, Epithelial Mesenchymal Transition)

|         |          |         |         |
|---------|----------|---------|---------|
| ABI3BP  | ACTA2    | ADAM12  | ANPEP   |
| APLP1   | AREG     | BASP1   | BDNF    |
| BGN     | BMP1     | CADM1   | CALD1   |
| CALU    | CAP2     | CAPG    | CCN1    |
| CCN2    | CD44     | CD59    | CDH11   |
| CDH2    | CDH6     | COL11A1 | COL12A1 |
| COL16A1 | COL1A1   | COL1A2  | COL3A1  |
| COL4A1  | COL4A2   | COL5A1  | COL5A2  |
| COL5A3  | COL6A2   | COL6A3  | COL7A1  |
| COL8A2  | COLGALT1 | COMP    | COPA    |
| CRLF1   | CTHRC1   | CXCL1   | CXCL12  |
| CXCL6   | CXCL8    | DAB2    | DCN     |
| DKK1    | DPYSL3   | DST     | ECM1    |
| ECM2    | EDIL3    | EFEMP2  | ELN     |
| EMP3    | ENO2     | FAP     | FAS     |
| FBLN1   | FBLN2    | FBLN5   | FBN1    |
| FBN2    | FERMT2   | FGF2    | FLNA    |
| FMOD    | FN1      | FOXC2   | FSTL1   |

|           |           |        |          |
|-----------|-----------|--------|----------|
| FSTL3     | FUCA1     | FZD8   | GADD45A  |
| GADD45B   | GAS1      | GEM    | GJA1     |
| GLIPR1    | GPC1      | GPX7   | GREM1    |
| HTRA1     | ID2       | IGFBP2 | IGFBP3   |
| IGFBP4    | IL15      | IL32   | IL6      |
| INHBA     | ITGA2     | ITGA5  | ITGAV    |
| ITGB1     | ITGB3     | ITGB5  | JUN      |
| LAMA1     | LAMA2     | LAMA3  | LAMC1    |
| LAMC2     | LGALS1    | LOX    | LOXL1    |
| LOXL2     | LRP1      | LRRC15 | LUM      |
| MAGEE1    | MATN2     | MATN3  | MCM7     |
| MEST      | MFAP5     | MGP    | MMP1     |
| MMP14     | MMP2      | MMP3   | MSX1     |
| MXRA5     | MYL9      | MYLK   | NID2     |
| NNMT      | NOTCH2    | NT5E   | NTM      |
| OXTR      | P3H1      | PCOLCE | PCOLCE2  |
| PDGFRB    | PDLIM4    | PFN2   | PLAUR    |
| PLOD1     | PLOD2     | PLOD3  | PMEPA1   |
| PMP22     | POSTN     | PIIB   | PRRX1    |
| PRSS2     | PTHLH     | PTX3   | PVR      |
| QSOX1     | RGS4      | RHOB   | SAT1     |
| SCG2      | SDC1      | SDC4   | SERPINE1 |
| SERPINE2  | SERPINH1  | SFRP1  | SFRP4    |
| SGCB      | SGCD      | SGCG   | SLC6A8   |
| SLIT2     | SLIT3     | SNAI2  | SNTB1    |
| SPARC     | SPOCK1    | SPP1   | TAGLN    |
| TFPI2     | TGFB1     | TGFB1  | TGFB1    |
| TGM2      | THBS1     | THBS2  | THY1     |
| TIMP1     | TIMP3     | TNC    | TNFAIP3  |
| TNFRSF11B | TNFRSF12A | TPM1   | TPM2     |
| TPM4      | VCAM1     | VCAN   | VEGFA    |
| VEGFC     | VIM       | WIPF1  | WNT5A    |

### **M3\_neuronal\_sensory\_pain (21 genes, Sensory Perception of Pain (GO:0019233))**

|          |         |        |         |
|----------|---------|--------|---------|
| CCL2     | CCR2    | CHRNA4 | CHRNA4  |
| KCNA2    | KCNK4   | MMP12  | MRGPRX2 |
| NIPSNAP1 | OPRK1   | OPRL1  | OPRM1   |
| P2RX4    | P2RX7   | PRDM12 | PROK2   |
| PTGES    | RETREG1 | SCN11A | SCN9A   |
| TRPA1    |         |        |         |

### **M4\_metabolism\_oxphos (200 genes, Oxidative Phosphorylation)**

|         |         |         |         |
|---------|---------|---------|---------|
| ABCB7   | ACAA1   | ACAA2   | ACADM   |
| ACADSB  | ACADVL  | ACAT1   | ACO2    |
| AFG3L2  | AIFM1   | ALAS1   | ALDH6A1 |
| ATP1B1  | ATP5F1A | ATP5F1B | ATP5F1C |
| ATP5F1D | ATP5F1E | ATP5MC1 | ATP5MC2 |
| ATP5MC3 | ATP5ME  | ATP5MF  | ATP5MG  |

|          |          |          |          |
|----------|----------|----------|----------|
| ATP5PB   | ATP5PD   | ATP5PF   | ATP5PO   |
| ATP6AP1  | ATP6V0B  | ATP6V0C  | ATP6V0E1 |
| ATP6V1C1 | ATP6V1D  | ATP6V1E1 | ATP6V1F  |
| ATP6V1G1 | ATP6V1H  | BAX      | BCKDHA   |
| BDH2     | CASP7    | COX10    | COX11    |
| COX15    | COX17    | COX4I1   | COX5A    |
| COX5B    | COX6A1   | COX6B1   | COX6C    |
| COX7A2   | COX7A2L  | COX7B    | COX7C    |
| COX8A    | CPT1A    | CS       | CYB5A    |
| CYB5R3   | CYC1     | CYCS     | DECR1    |
| DLAT     | DLD      | DLST     | ECH1     |
| ECHS1    | ECI1     | ETFA     | ETFB     |
| ETFDH    | FDX1     | FH       | FXN      |
| GLUD1    | GOT2     | GPI      | GPX4     |
| GRPEL1   | HADHA    | HADHB    | HCCS     |
| HSD17B10 | HSPA9    | HTRA2    | IDH1     |
| IDH2     | IDH3A    | IDH3B    | IDH3G    |
| IMMT     | ISCA1    | ISCU     | LDHA     |
| LDHB     | LRPPRC   | MAOB     | MDH1     |
| MDH2     | MFN2     | MGST3    | MPC1     |
| MRPL11   | MRPL15   | MRPL34   | MRPL35   |
| MRPS11   | MRPS12   | MRPS15   | MRPS22   |
| MRPS30   | MTRF1    | MTRR     | MTX2     |
| NDUFA1   | NDUFA2   | NDUFA3   | NDUFA4   |
| NDUFA5   | NDUFA6   | NDUFA7   | NDUFA8   |
| NDUFA9   | NDUFAB1  | NDUFB1   | NDUFB2   |
| NDUFB3   | NDUFB4   | NDUFB5   | NDUFB6   |
| NDUFB7   | NDUFB8   | NDUFC1   | NDUFC2   |
| NDUFS1   | NDUFS2   | NDUFS3   | NDUFS4   |
| NDUFS6   | NDUFS7   | NDUFS8   | NDUFV1   |
| NDUFV2   | NNT      | NQO2     | OAT      |
| OGDH     | OPA1     | OXA1L    | PDHA1    |
| PDHB     | PDHX     | PDK4     | PDP1     |
| PHB2     | PHYH     | PMPCA    | POLR2F   |
| POR      | PRDX3    | RETSAT   | RHOT1    |
| RHOT2    | SDHA     | SDHB     | SDHC     |
| SDHD     | SLC25A11 | SLC25A12 | SLC25A20 |
| SLC25A3  | SLC25A4  | SLC25A5  | SLC25A6  |
| SUCLA2   | SUCLG1   | SUPV3L1  | SURF1    |
| TCIRG1   | TIMM10   | TIMM13   | TIMM17A  |
| TIMM50   | TIMM8B   | TIMM9    | TOMM22   |
| TOMM70   | UQCR10   | UQCR11   | UQCRB    |
| UQCRC1   | UQCRC2   | UQCRFS1  | UQCRH    |
| UQCRQ    | VDAC1    | VDAC2    | VDAC3    |

#### **M5\_stress\_repair\_hypoxia (200 genes, Hypoxia)**

|        |         |         |          |
|--------|---------|---------|----------|
| ACKR3  | ADM     | ADORA2B | AK4      |
| AKAP12 | ALDOA   | ALDOB   | ALDOC    |
| AMPD3  | ANGPTL4 | ANKZF1  | ANXA2    |
| ATF3   | ATP7A   | B3GALT6 | B4GALNT2 |

|          |         |          |          |
|----------|---------|----------|----------|
| BCAN     | BCL2    | BGN      | BHLHE40  |
| BNIP3L   | BRS3    | BTG1     | CA12     |
| CASP6    | CAV1    | CAVIN1   | CAVIN3   |
| CCN1     | CCN2    | CCN5     | CCNG2    |
| CDKN1A   | CDKN1B  | CDKN1C   | CHST2    |
| CHST3    | CITED2  | COL5A1   | CP       |
| CSRP2    | CXCR4   | DCN      | DDIT3    |
| DDIT4    | DPYSL4  | DTNA     | DUSP1    |
| EDN2     | EFNA1   | EFNA3    | EGFR     |
| ENO1     | ENO2    | ENO3     | ERO1A    |
| ERRFI1   | ETS1    | EXT1     | F3       |
| FAM162A  | FBP1    | FOS      | FOSL2    |
| FOXO3    | GAA     | GALK1    | GAPDH    |
| GAPDHS   | GBE1    | GCK      | GCNT2    |
| GLRX     | GPC1    | GPC3     | GPC4     |
| GPI      | GRHR    | GYS1     | HAS1     |
| HDLBP    | HEXA    | HK1      | HK2      |
| HMOX1    | HOXB9   | HS3ST1   | HSPA5    |
| IDS      | IER3    | IGFBP1   | IGFBP3   |
| IL6      | ILVBL   | INHA     | IRS2     |
| ISG20    | JMJD6   | JUN      | KDELRL3  |
| KDM3A    | KIF5A   | KLF6     | KLF7     |
| KLHL24   | LALBA   | LARGE1   | LDHA     |
| LDHC     | LOX     | LXN      | MAFF     |
| MAP3K1   | MIF     | MT1E     | MT2A     |
| MXI1     | MYH9    | NAGK     | NCAN     |
| NDRG1    | NDST1   | NDST2    | NEDD4L   |
| NFIL3    | NOCT    | NR3C1    | P4HA1    |
| P4HA2    | PAM     | PCK1     | PDGFB    |
| PDK1     | PDK3    | PFKFB3   | PFKL     |
| PFKP     | PGAM2   | PGF      | PGK1     |
| PGM1     | PGM2    | PHKG1    | PIM1     |
| PKLR     | PKP1    | PLAC8    | PLAUR    |
| PLIN2    | PNRC1   | PPARGC1A | PPFIA4   |
| PPP1R15A | PPP1R3C | PRDX5    | PRKCA    |
| PYGM     | RBPJ    | RORA     | RRAGD    |
| S100A4   | SAP30   | SCARB1   | SDC2     |
| SDC3     | SDC4    | SELENBP1 | SERPINE1 |
| SIAH2    | SLC25A1 | SLC2A1   | SLC2A3   |
| SLC2A5   | SLC37A4 | SLC6A6   | SRPX     |
| STBD1    | STC1    | STC2     | SULT2B1  |
| TES      | TGFB3   | TGFBI    | TGM2     |
| TIPARP   | TKTL1   | TMEM45A  | TNFAIP3  |
| TPBG     | TPD52   | TPI1     | TPST2    |
| UGP2     | VEGFA   | VHL      | VLDLR    |
| WSB1     | XPNPEP1 | ZFP36    | ZNF292   |

#### M6\_cell\_state\_g2m (200 genes, G2-M Checkpoint)

|      |       |        |       |
|------|-------|--------|-------|
| ABL1 | AMD1  | ARID4A | ATF5  |
| ATRX | AURKA | AURKB  | BARD1 |

|          |          |         |         |
|----------|----------|---------|---------|
| BCL3     | BIRC5    | BRCA2   | BUB1    |
| BUB3     | CASP8AP2 | CBX1    | CCNA2   |
| CCNB2    | CCND1    | CCNF    | CCNT1   |
| CDC20    | CDC25A   | CDC25B  | CDC27   |
| CDC45    | CDC6     | CDC7    | CDK1    |
| CDK4     | CDKN1B   | CDKN2C  | CDKN3   |
| CENPA    | CENPE    | CENPF   | CHAF1A  |
| CHEK1    | CHMP1A   | CKS1B   | CKS2    |
| CTCF     | CUL1     | CUL3    | CUL4A   |
| CUL5     | DBF4     | DDX39A  | DKC1    |
| DMD      | DR1      | DTYMK   | E2F1    |
| E2F2     | E2F3     | E2F4    | EFNA5   |
| EGF      | ESPL1    | EWSR1   | EXO1    |
| EZH2     | FANCC    | FBXO5   | FOXN3   |
| G3BP1    | GINS2    | GSPT1   | H2AX    |
| H2AZ1    | H2AZ2    | H2BC12  | HIF1A   |
| HIRA     | HMGA1    | HMGB3   | HMGN2   |
| HMMR     | HNRNPD   | HNRNPU  | HOXC10  |
| HSPA8    | HUS1     | ILF3    | INCENP  |
| JPT1     | KATNA1   | KIF11   | KIF15   |
| KIF20B   | KIF22    | KIF23   | KIF2C   |
| KIF4A    | KIF5B    | KMT5A   | KNL1    |
| KPNA2    | KPNB1    | LBR     | LIG3    |
| LMNB1    | MAD2L1   | MAP3K20 | MAPK14  |
| MARCKS   | MCM2     | MCM3    | MCM5    |
| MCM6     | MEIS1    | MEIS2   | MKI67   |
| MNAT1    | MT2A     | MTF2    | MYBL2   |
| MYC      | NASP     | NCL     | NDC80   |
| NEK2     | NOLC1    | NOTCH2  | NSD2    |
| NUMA1    | NUP50    | NUP98   | NUSAP1  |
| ODC1     | ODF2     | ORC5    | ORC6    |
| PAFAH1B1 | PBK      | PDS5B   | PLK1    |
| PLK4     | PML      | POLA2   | POLE    |
| POLQ     | PRC1     | PRIM2   | PRMT5   |
| PRPF4B   | PTTG1    | PTTG3P  | PURA    |
| RACGAP1  | RAD21    | RAD23B  | RAD54L  |
| RASAL2   | RBL1     | RBM14   | RPA2    |
| RPS6KA5  | SAP30    | SFPQ    | SLC12A2 |
| SLC38A1  | SLC7A1   | SLC7A5  | SMAD3   |
| SMARCC1  | SMC1A    | SMC2    | SMC4    |
| SNRPD1   | SQLE     | SRSF1   | SRSF10  |
| SRSF2    | SS18     | STAG1   | STIL    |
| STMN1    | SUV39H1  | SYNCRIP | TACC3   |
| TENT4A   | TFDP1    | TGFB1   | TLE3    |
| TMPO     | TNPO2    | TOP1    | TOP2A   |
| TPX2     | TRA2B    | TRAIP   | TROAP   |
| TTK      | UBE2C    | UBE2S   | UCK2    |
| UPF1     | WRN      | XPO1    | YTHDC1  |

## D Pre-specified miRNA regulators (A2.3b)

Named human regulators of the M6 module from miRTarBase v10 (87,627 validated pairs); 262 mature miRNAs measured in GSE196156.

|                         |                         |
|-------------------------|-------------------------|
| hsa-let-7-5p/98-5p      | hsa-let-7a-2-3p         |
| hsa-let-7a-5p           | hsa-let-7b-3p           |
| hsa-let-7d-3p           | hsa-let-7d-5p           |
| hsa-let-7f-2-3p         | hsa-let-7f-2-3p/1185-3p |
| hsa-let-7i-3p           | hsa-let-7i-5p           |
| hsa-miR-10-5p           | hsa-miR-100-3p          |
| hsa-miR-101-3p.1        | hsa-miR-101-3p.2        |
| hsa-miR-103a-1-5p       | hsa-miR-103a-2-5p       |
| hsa-miR-105-3p          | hsa-miR-105-5p          |
| hsa-miR-106a-5p         | hsa-miR-106b-3p         |
| hsa-miR-10a-5p          | hsa-miR-10b-3p          |
| hsa-miR-1179            | hsa-miR-1180-3p         |
| hsa-miR-1184            | hsa-miR-1185-1-3p       |
| hsa-miR-1197            | hsa-miR-1199-3p         |
| hsa-miR-1203            | hsa-miR-1204            |
| hsa-miR-1207-5p         | hsa-miR-1207-5p/4763-3p |
| hsa-miR-1224-3p         | hsa-miR-1224-5p         |
| hsa-miR-1226-5p         | hsa-miR-1227-3p         |
| hsa-miR-1229-3p         | hsa-miR-1229-5p         |
| hsa-miR-1233-5p/6778-5p | hsa-miR-1234-3p         |
| hsa-miR-1237-3p         | hsa-miR-1237-5p         |
| hsa-miR-124-3p          | hsa-miR-124-3p.1        |
| hsa-miR-1244            | hsa-miR-1245a           |
| hsa-miR-1246            | hsa-miR-1247-3p         |
| hsa-miR-1249-5p         | hsa-miR-1249-5p/6797-5p |
| hsa-miR-1251-3p         | hsa-miR-1251-5p         |
| hsa-miR-1254/3116       | hsa-miR-1255a           |
| hsa-miR-1256            | hsa-miR-1257            |
| hsa-miR-125b-1-3p       | hsa-miR-125b-2-3p       |
| hsa-miR-126-3p.2        | hsa-miR-126-5p          |
| hsa-miR-1261            | hsa-miR-1262            |
| hsa-miR-1265            | hsa-miR-1266-3p         |
| hsa-miR-1268a/1268b     | hsa-miR-1269            |
| hsa-miR-127-5p          | hsa-miR-1270            |
| hsa-miR-1273a           | hsa-miR-1273c           |
| hsa-miR-1273g-3p        | hsa-miR-1273g-5p        |
| hsa-miR-1275/4665-5p    | hsa-miR-1276            |
| hsa-miR-1279            | hsa-miR-128-1-5p        |
| hsa-miR-1283            | hsa-miR-1284            |
| hsa-miR-1287-3p         | hsa-miR-1287-5p         |
| hsa-miR-128b            | hsa-miR-129-1-3p        |
| hsa-miR-1290            | hsa-miR-1291/6775-3p    |
| hsa-miR-1293/4483       | hsa-miR-1294            |
| hsa-miR-1295b-5p/1912   | hsa-miR-1296-3p         |
| hsa-miR-1298-5p         | hsa-miR-1299            |
| hsa-miR-1302            | hsa-miR-1302/4298       |
| hsa-miR-1305            | hsa-miR-1306-3p         |
| hsa-miR-130a-3p         | hsa-miR-130a-5p         |

|                                         |                       |
|-----------------------------------------|-----------------------|
| hsa-miR-132-3p/212-3p                   | hsa-miR-132-5p        |
| hsa-miR-1324                            | hsa-miR-133a-3p       |
| hsa-miR-134-3p                          | hsa-miR-134-5p        |
| hsa-miR-135a-3p                         | hsa-miR-135a-5p       |
| hsa-miR-136-5p                          | hsa-miR-137           |
| hsa-miR-138-5p                          | hsa-miR-139-3p        |
| hsa-miR-140-3p.2                        | hsa-miR-140-5p        |
| hsa-miR-142-3p                          | hsa-miR-142-3p.1      |
| hsa-miR-143-5p                          | hsa-miR-144-3p        |
| hsa-miR-146-5p                          | hsa-miR-1468-3p       |
| hsa-miR-146a-5p                         | hsa-miR-146b-3p       |
| hsa-miR-147a                            | hsa-miR-147b          |
| hsa-miR-148a-5p                         | hsa-miR-148b-3p       |
| hsa-miR-149-3p/4728-5p/6785-5p/6883-5p  | hsa-miR-149-5p        |
| hsa-miR-151-3p                          | hsa-miR-151-5p        |
| hsa-miR-152-5p                          | hsa-miR-153-3p        |
| hsa-miR-1538                            | hsa-miR-1538/4745-3p  |
| hsa-miR-155-3p                          | hsa-miR-155-5p        |
| hsa-miR-15a-5p                          | hsa-miR-15b-3p        |
| hsa-miR-16-2-3p/195-3p                  | hsa-miR-16-5p         |
| hsa-miR-17-5p/20-5p/93-5p/106-5p/519-3p | hsa-miR-18-5p         |
| hsa-miR-181a-3p                         | hsa-miR-181a-5p       |
| hsa-miR-181d-3p                         | hsa-miR-181d-5p       |
| hsa-miR-1827                            | hsa-miR-183-3p        |
| hsa-miR-184                             | hsa-miR-1843          |
| hsa-miR-186-5p                          | hsa-miR-187-3p        |
| hsa-miR-18a-3p                          | hsa-miR-18a-5p        |
| hsa-miR-19-5p                           | hsa-miR-190-5p        |
| hsa-miR-1909-3p/6722-3p                 | hsa-miR-1909-5p       |
| hsa-miR-191-3p                          | hsa-miR-191-5p        |
| hsa-miR-1911-3p                         | hsa-miR-1911-5p       |
| hsa-miR-1914-5p                         | hsa-miR-1915-3p       |
| hsa-miR-192-5p                          | hsa-miR-192-5p/215-5p |
| hsa-miR-193b-3p                         | hsa-miR-193b-5p       |
| hsa-miR-195-5p                          | hsa-miR-196-5p        |
| hsa-miR-196b-5p                         | hsa-miR-197-3p        |
| hsa-miR-1973                            | hsa-miR-1976          |
| hsa-miR-199a-3p                         | hsa-miR-199a-5p       |
| hsa-miR-19a-5p                          | hsa-miR-19b-1-5p      |
| hsa-miR-200a-5p                         | hsa-miR-200ab-5p      |
| hsa-miR-200c-3p                         | hsa-miR-200c-5p       |
| hsa-miR-203a-3p                         | hsa-miR-203a-3p.1     |
| hsa-miR-203b-5p/6718-5p                 | hsa-miR-204-3p        |
| hsa-miR-205-3p                          | hsa-miR-205-5p        |
| hsa-miR-206                             | hsa-miR-208-3p        |
| hsa-miR-20a-3p                          | hsa-miR-20a-5p        |
| hsa-miR-21-3p/3591-3p                   | hsa-miR-21-5p         |
| hsa-miR-211-3p                          | hsa-miR-211-5p        |
| hsa-miR-2114-5p                         | hsa-miR-2115-3p       |
| hsa-miR-2117                            | hsa-miR-212-3p        |
| hsa-miR-214-5p                          | hsa-miR-215-3p        |
| hsa-miR-216b-3p                         | hsa-miR-216b-5p       |

|                              |                                         |
|------------------------------|-----------------------------------------|
| hsa-miR-218-5p               | hsa-miR-219-5p                          |
| hsa-miR-219b-3p              | hsa-miR-219b-5p                         |
| hsa-miR-221-3p/222-3p        | hsa-miR-221-5p                          |
| hsa-miR-223-3p               | hsa-miR-223-5p                          |
| hsa-miR-2276-5p              | hsa-miR-2277-3p                         |
| hsa-miR-2355-3p              | hsa-miR-2355-5p                         |
| hsa-miR-23a-5p               | hsa-miR-23b-3p                          |
| hsa-miR-24-2-5p              | hsa-miR-24-3p                           |
| hsa-miR-25-3p                | hsa-miR-25-3p/32-5p/92-3p/363-3p/367-3p |
| hsa-miR-2681-3p              | hsa-miR-2681-5p                         |
| hsa-miR-26a-1-3p             | hsa-miR-26a-2-3p                        |
| hsa-miR-27-3p                | hsa-miR-27a-3p                          |
| hsa-miR-28-3p                | hsa-miR-28-5p                           |
| hsa-miR-2909                 | hsa-miR-296-3p                          |
| hsa-miR-299-3p               | hsa-miR-299-5p                          |
| hsa-miR-29b-2-5p             | hsa-miR-29b-3p                          |
| hsa-miR-30-5p                | hsa-miR-300                             |
| hsa-miR-301b-3p              | hsa-miR-301b-5p                         |
| hsa-miR-302b-3p              | hsa-miR-302b-5p                         |
| hsa-miR-302c-5p              | hsa-miR-302d-3p                         |
| hsa-miR-3064-3p              | hsa-miR-3064-5p                         |
| hsa-miR-3074-5p              | hsa-miR-30a-3p                          |
| hsa-miR-30b-5p               | hsa-miR-30c-1-3p                        |
| hsa-miR-30d-3p               | hsa-miR-30d-5p                          |
| hsa-miR-31-5p                | hsa-miR-3115                            |
| hsa-miR-3119                 | hsa-miR-3120-3p                         |
| hsa-miR-3122                 | hsa-miR-3122/3913-5p                    |
| hsa-miR-3125/3916/6859-5p    | hsa-miR-3126-3p                         |
| hsa-miR-3127-5p              | hsa-miR-3128                            |
| hsa-miR-3130-3p              | hsa-miR-3130-5p                         |
| hsa-miR-3134                 | hsa-miR-3135a                           |
| hsa-miR-3136-5p              | hsa-miR-3136-5p/4439                    |
| hsa-miR-3140-3p              | hsa-miR-3140-5p                         |
| hsa-miR-3144-5p              | hsa-miR-3145-3p                         |
| hsa-miR-3148                 | hsa-miR-3149                            |
| hsa-miR-3151-3p              | hsa-miR-3151-5p                         |
| hsa-miR-3153/6733-5p/6739-5p | hsa-miR-3154                            |
| hsa-miR-3156-5p              | hsa-miR-3157-3p                         |
| hsa-miR-3159                 | hsa-miR-3160-3p                         |
| hsa-miR-3162-5p              | hsa-miR-3163                            |
| hsa-miR-3167                 | hsa-miR-3168                            |
| hsa-miR-3171                 | hsa-miR-3173-3p                         |
| hsa-miR-3175                 | hsa-miR-3176                            |
| hsa-miR-3178                 | hsa-miR-3179                            |
| hsa-miR-3181                 | hsa-miR-3182                            |
| hsa-miR-3185                 | hsa-miR-3186-3p                         |
| hsa-miR-3189-3p              | hsa-miR-3189-5p                         |
| hsa-miR-3191-5p              | hsa-miR-3192-3p                         |
| hsa-miR-3194-5p              | hsa-miR-3195                            |
| hsa-miR-3198/4309            | hsa-miR-3199                            |
| hsa-miR-320                  | hsa-miR-3200-3p                         |
| hsa-miR-3202                 | hsa-miR-320a                            |

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| hsa-miR-6892-5p | hsa-miR-6893-3p             |
| hsa-miR-6895-3p | hsa-miR-6895-5p             |
| hsa-miR-7-5p    | hsa-miR-708-3p              |
| hsa-miR-7107-5p | hsa-miR-7108-3p             |
| hsa-miR-711     | hsa-miR-7110-3p             |
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| hsa-miR-7158-5p | hsa-miR-7159-3p             |
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| hsa-miR-802     | hsa-miR-8052                |
| hsa-miR-8056    | hsa-miR-8057                |
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| hsa-miR-943     | hsa-miR-944                 |
| hsa-miR-96-3p   | hsa-miR-96-5p               |
| hsa-miR-99-3p   | hsa-miR-99-5p/100-5p        |
| hsa-miR-99b-3p  | hsa-miR-99b-5p              |

## E External-service ledger

Counting conventions and the rejected candidates with evidence are in `EXTERNAL_SERVICES.md`; dual count: 43 counted / 41 most-conservative.

| #  | Service                                                 | Genuine use in this lane                                                                                                                                                     | Evidence                                                      |
|----|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| 1  | NCBI GEO (search/FTP/SOFT)                              | Accession discovery, series/sample SOFT retrieval, matrix download for GSE238208, GSE196156, GSE11839, GSE28242, GSE57560, GSE621, GSE11783                                  | sources/*.txt, sources/*_crosswalk.csv                        |
| 2  | NCBI GEO platform tables (GPL6244, GPL29077 annotation) | Symbol mapping for urine transfer (190/190 M6 symbols)                                                                                                                       | results/ic_p48_a2_urine_gse28242_result.js                    |
| 3  | Enrichr (Maayan Lab) - MSigDB_Hallmark_2020 library     | M0-M6 module gene sets (locked)                                                                                                                                              | prereg/ic_p48_module_genesets.json                            |
| 4  | Enrichr - GO_Biological_Process_2025 library            | M3 pain module gene set (GO:0019233, 21 genes, locked)                                                                                                                       | prereg/ic_p48_module_genesets.json                            |
| 5  | ChatGPT (program consult, redirect mode)                | Round 1 redirect after corrected k=2 negative; round 2 residual-framework consult                                                                                            | redirect/round1_*, redirect/round2_*                          |
| 6  | miRTarBase v10 (new host awi.cuhk.edu.cn)               | Human validated miRNA-target pairs vs the 200-gene M6 module (87,627 pairs; per-gene CSV export endpoint)                                                                    | results/ic_p48_a2_urine_gse196156_result.js                   |
| 7  | miRBase (mature.fa)                                     | MIMAT accession -i mature miRNA name map (2,789 hsa entries) for A2.3b                                                                                                       | results/ic_p48_a2_urine_gse196156_result.js                   |
| 8  | Ensembl REST                                            | Per-gene lookup for M6 module genes                                                                                                                                          | projects/interstitial_cystitis/results/external               |
| 9  | UniProtKB REST                                          | Reviewed human accessions per M6 gene                                                                                                                                        | same                                                          |
| 10 | Europe PMC                                              | "interstitial cystitis" co-mention literature per M6 gene                                                                                                                    | same                                                          |
| 11 | MyGene.info                                             | Identifier cross-check per M6 gene                                                                                                                                           | same                                                          |
| 12 | ChEMBL API                                              | Target lookup per M6 gene                                                                                                                                                    | same                                                          |
| 13 | HGNC REST (gene-names.org)                              | Symbol status/ID per M6 gene                                                                                                                                                 | same                                                          |
| 14 | GTEx Portal API                                         | Reference gene lookup per M6 gene                                                                                                                                            | same                                                          |
| 15 | Reactome Content Service                                | Pathway mapping via UniProt accession per M6 gene                                                                                                                            | same                                                          |
| 16 | g:Profiler g:GOST                                       | Enrichment of the committed 200-gene M6 list: top hits mitotic cell cycle (GO:1903047/GO:0000278/GO:0007049) and REAC Cell Cycle - operational validation of the module list | results/external/secondary_annotations_m6 (in this directory) |
| 17 | STRING API                                              | Network query among the 5 miRNA-hub M6 genes: 0 physical edges returned (honest null: miRNA hubs are not a physical complex)                                                 | same                                                          |
| 18 | Open Targets Platform GraphQL                           | Disease search resolved interstitial cystitis = EFO_1000869; hub genes G3BP1/SRSF1/DR1 not among its top-50 associated targets (honest null, recorded)                       | same                                                          |
| 19 | EMBL-EBI (MONDO)                                        | OLS Disease ontology anchor MONDO:0018301 "interstitial cystitis"                                                                                                            | same                                                          |
| 20 | QuickGO                                                 | GO annotations per hub gene                                                                                                                                                  | same                                                          |
| 21 | IntAct                                                  | Interaction records per hub gene                                                                                                                                             | same                                                          |
| 22 | InterPro                                                | Domain annotations per hub gene                                                                                                                                              | same                                                          |

| #  | Service                    | Genuine use in this lane                                                                                                                                                        | Evidence                                      |
|----|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 23 | OpenAlex                   | "hub gene + interstitial cystitis" literature metadata                                                                                                                          | same                                          |
| 24 | Human Protein Atlas        | Per-gene tissue/protein context (via Ensembl IDs)                                                                                                                               | same                                          |
| 25 | DGIdb (GraphQL)            | Drug-gene interactions per hub: MAPK14 druggable (105 interactions incl. p38 inhibitors); other hubs 0 - honest druggability gradient                                           | results/external/tertiary_annotations_m6.js   |
| 26 | GWAS Catalog REST v2       | Mapped-gene associations per hub (verified 'mapped_gene=' filter with a bogus-gene control; traits recorded per gene)                                                           | same                                          |
| 27 | NCBI PubMed eutils         | Programmatic "interstitial cystitis" Title/Abstract co-mention counts per hub: all 0 - honest null (IC abstracts do not co-mention these hubs)                                  | same                                          |
| 28 | Pharos (NIH/NCATS GraphQL) | Target development levels: MAPK14 Tchem/Kinase; G3BP1, DR1, RASAL2 Tbio; SRSF1 Tbio                                                                                             | same                                          |
| 29 | cBioPortal API             | Hub mutation records in TCGA bladder (blca_tcga_pan_can_atlas_2018): RASAL2 16, SRSF1 6, MAPK14 4, G3BP1 2, DR1 1                                                               | same                                          |
| 30 | WikiPathways JSON API      | Human pathway search per hub (e.g. MAPK14 in 1,131 human pathway text hits)                                                                                                     | same                                          |
| 31 | PGS Catalog REST           | Trait search "cystitis"/"bladder pain syndrome": 0 PGS traits - honest empty                                                                                                    | same                                          |
| 32 | Monarch Initiative v3      | IC disease record (MONDO:0018301) + association predicates + disease phenotypes (bladder/urethra/pain features recorded)                                                        | results/external/quaternary_annotations_m6.js |
| 33 | NCBI ClinVar (eutils)      | Clinical variant record counts per hub (RASAL2 222 ... DR1 20); distinct NCBI database, GEO-vs-PubMed counting precedent                                                        | same                                          |
| 34 | NCBI dbSNP (eutils)        | Human variant counts per hub (RASAL2 145,551 ... SRSF1 9,441); distinct NCBI database                                                                                           | same                                          |
| 35 | NCBI Gene (eutils)         | Gene records/summaries per hub; distinct NCBI database                                                                                                                          | same                                          |
| 36 | UCSC Genome Browser API    | hg38 locus search per hub (position matches recorded)                                                                                                                           | same                                          |
| 37 | Signor                     | Full-network download (43,570 rows) filtered to human hub edges: MAPK14 239, G3BP1 11, SRSF1 9, DR1 1, RASAL2 0 (per-protein params verified ignored; network filtered locally) | same                                          |
| 38 | KEGG REST                  | Gene records + pathway membership per hub (MAPK14 hsa:1432 in 65 pathways; G3BP1/SRSF1/RASAL2 sparse - honest gradient)                                                         | results/external/quinary_annotations_m6.js    |

| #  | Service                              | Genuine use in this lane                                                                                                         | Evidence                                    |
|----|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| 39 | GO Consortium Central API            | Term records behind locked modules: GO:0000086 G2/M transition (M6), GO:0019233 sensory perception of pain (M3)                  | same                                        |
| 40 | Rfam                                 | Family records for the top-3 A2.3b regulator miRNAs by validated M6 pair count: mir-16 RF00254, mir-24 RF00178, mir-3613 RF03323 | same                                        |
| 41 | RNAcentral                           | RNA record for top regulator hsa-miR-16-5p (URS00004BCD9C)                                                                       | same                                        |
| 42 | Pathway Commons (NCI-PID datasource) | Pathway search per hub restricted to NCI-PID (Reactome excluded to avoid alias): MAPK14 33 hits (top: p38-alpha/beta signaling)  | results/external/buffer_annotations_m6.json |
| 43 | ClinicalTrials.gov API v2            | IC/BPS trial landscape sample + p38 MAPK inhibitor trial search (context for the DGIdb druggability finding)                     | same                                        |

## F Per-gene external annotation digest

One line per M6 module gene, from the committed per-gene snapshots; columns are counts returned by each service, not curated judgments.

| Gene     | UniProt | EuropePMC-IC hits | ChEMBL targets | Reactome pathways |
|----------|---------|-------------------|----------------|-------------------|
| ABL1     | P00519  | 13                | 17             | 13                |
| AMD1     | P17707  | 0                 | 3              | 1                 |
| ARID4A   | P29374  | 0                 | 1              | 2                 |
| ATF5     | Q9Y2D1  | 4                 | 0              | 2                 |
| ATRX     | P46100  | 8                 | 1              | 3                 |
| AURKA    | O14965  | 12                | 10             | 8                 |
| AURKB    | Q96GD4  | 3                 | 8              | 10                |
| BARD1    | Q99728  | 2                 | 0              | 21                |
| BCL3     | P20749  | 3                 | 1              |                   |
| BIRC5    | O15392  | 9                 | 1              | 10                |
| BRCA2    | P51587  | 32                | 2              | 14                |
| BUB1     | O43683  | 3                 | 2              | 6                 |
| BUB3     | O43684  | 1                 | 2              | 10                |
| CASP8AP2 | Q9UKL3  | 0                 | 0              | 1                 |
| CBX1     | P83916  | 1                 | 1              | 2                 |
| CCNA2    | P20248  | 6                 | 8              | 21                |
| CCNB2    | O95067  | 5                 | 2              | 11                |
| CCND1    | P24385  | 28                | 10             | 18                |
| CCNF     | P41002  | 0                 | 0              | 2                 |
| CCNT1    | O60563  | 1                 | 5              | 16                |
| CDC20    | Q12834  | 3                 | 2              | 20                |
| CDC25A   | P30304  | 7                 | 1              | 9                 |
| CDC25B   | P30305  | 2                 | 2              | 4                 |
| CDC27    | P30260  | 1                 | 3              | 18                |
| CDC45    | O75419  | 1                 | 1              | 4                 |
| CDC6     | Q99741  | 3                 | 1              | 8                 |

| Gene   | UniProt | EuropePMC-IC hits | ChEMBL targets | Reactome pathways |
|--------|---------|-------------------|----------------|-------------------|
| CDC7   | O00311  | 6                 | 4              | 3                 |
| CDK1   | P06493  | 31                | 17             | 36                |
| CDK4   | P11802  | 25                | 24             | 19                |
| CDKN1B | P46527  | 10                | 2              | 16                |
| CDKN2C | P42773  | 2                 | 0              | 4                 |
| CDKN3  | Q00526  | 0                 | 6              |                   |
| CENPA  | P49450  | 1                 | 0              | 7                 |
| CENPE  | Q02224  | 0                 | 8              | 10                |
| CENPF  | P49454  | 4                 | 0              | 7                 |
| CHAF1A | Q13111  | 0                 | 0              |                   |
| CHEK1  | O14757  | 4                 | 2              | 10                |
| CHMP1A | Q9HD42  | 0                 | 1              | 1                 |
| CKS1B  | P61024  | 0                 | 3              | 2                 |
| CKS2   | P33552  | 3                 | 1              |                   |
| CTCF   | P49711  | 9                 | 1              | 2                 |
| CUL1   | Q13616  | 4                 | 3              | 36                |
| CUL3   | Q13618  | 7                 | 4              | 11                |
| CUL4A  | Q13619  | 2                 | 10             | 10                |
| CUL5   | Q93034  | 2                 | 1              | 6                 |
| DBF4   | Q9UBU7  | 0                 | 2              | 2                 |
| DDX39A | O00148  | 0                 | 1              | 2                 |
| DKC1   | O60832  | 0                 | 1              | 2                 |
| DMD    | P11532  | 27                | 2              | 3                 |
| DR1    | Q01658  | 5                 | 1              | 2                 |
| DTYMK  | P23919  | 0                 | 1              | 1                 |
| E2F1   | Q01094  | 30                | 3              | 17                |
| E2F2   | Q14209  | 3                 | 1              | 5                 |
| E2F3   | O00716  | 5                 | 1              | 6                 |
| E2F4   | Q16254  | 1                 | 1              | 10                |
| EFNA5  | P52803  | 1                 | 0              | 3                 |
| EGF    | P01133  | 297               | 144            | 37                |
| ESPL1  | Q14674  | 1                 | 1              | 1                 |
| EWSR1  | Q01844  | 6                 | 2              |                   |
| EXO1   | Q9UQ84  | 0                 | 1              | 17                |
| EZH2   | Q15910  | 39                | 11             | 10                |
| FANCC  | Q00597  | 1                 | 1              | 3                 |
| FBXO5  | Q9UKT4  | 0                 | 0              | 5                 |
| FOXN3  | O00409  | 1                 | 0              |                   |
| G3BP1  | Q13283  | 3                 | 1              | 1                 |
| GINS2  | Q9Y248  | 0                 | 0              | 1                 |
| GSPT1  | P15170  | 3                 | 7              | 4                 |
| H2AX   | P16104  | 26                | 0              | 52                |
| H2AZ1  | P0C0S5  | 0                 | 1              | 2                 |
| H2AZ2  | Q71UI9  | 0                 | 0              | 45                |
| H2BC12 | O60814  | 0                 | 0              | 58                |
| HIF1A  | Q16665  | 267               | 6              | 11                |
| HIRA   | P54198  | 15                | 6              | 2                 |
| HMGA1  | P17096  | 6                 | 1              | 7                 |
| HMGB3  | O15347  | 0                 | 1              |                   |
| HMG2   | P05204  | 1                 | 0              |                   |
| HMMR   | O75330  | 3                 | 1              | 3                 |

| Gene    | UniProt | EuropePMC-IC hits | ChEMBL targets | Reactome pathways |
|---------|---------|-------------------|----------------|-------------------|
| HNRNPD  | Q14103  | 0                 | 1              | 6                 |
| HNRNPU  | Q00839  | 2                 | 1              | 5                 |
| HOXC10  | Q9NYD6  | 2                 | 0              |                   |
| HSPA8   | P11142  | 4                 | 3              | 21                |
| HUS1    | O60921  | 0                 | 0              | 7                 |
| ILF3    | Q12906  | 3                 | 1              | 2                 |
| INCENP  | Q9NQS7  | 5                 | 3              | 7                 |
| JPT1    | Q9UK76  | 0                 | 0              |                   |
| KATNA1  | O75449  | 0                 | 1              |                   |
| KIF11   | P52732  | 2                 | 4              | 4                 |
| KIF15   | Q9NS87  | 1                 | 1              | 4                 |
| KIF20B  | Q96Q89  | 0                 | 2              | 2                 |
| KIF22   | Q14807  | 1                 | 1              | 4                 |
| KIF23   | Q02241  | 2                 | 1              | 5                 |
| KIF2C   | Q99661  | 2                 | 1              | 10                |
| KIF4A   | O95239  | 1                 | 1              | 5                 |
| KIF5B   | P33176  | 6                 | 4              | 7                 |
| KMT5A   | Q9NQR1  | 0                 | 2              | 4                 |
| KNL1    | Q8NG31  | 3                 | 0              | 7                 |
| KPNA2   | P52292  | 3                 | 3              | 9                 |
| KPNB1   | Q14974  | 2                 | 3              | 15                |
| LBR     | Q14739  | 11                | 1              | 11                |
| LIG3    | P49916  | 0                 | 1              | 6                 |
| LMNB1   | P20700  | 2                 | 1              | 10                |
| MAD2L1  | Q13257  | 0                 | 0              | 11                |
| MAP3K20 | Q9NYL2  | 0                 | 2              |                   |
| MAPK14  | Q16539  | 6                 | 10             | 18                |
| MARCKS  | P29966  | 3                 | 1              | 1                 |
| MCM2    | P33993  | 6                 | 3              | 6                 |
| MCM3    | P25205  | 0                 | 2              | 6                 |
| MCM5    | P33992  | 3                 | 1              | 7                 |
| MCM6    | Q14566  | 2                 | 1              | 6                 |
| MEIS1   | O00470  | 5                 | 1              | 1                 |
| MEIS2   | O14770  | 2                 | 0              |                   |
| MKI67   | P46013  | 14                | 2              |                   |
| MNAT1   | P51948  | 1                 | 2              | 33                |
| MT2A    | P02795  | 2                 | 0              | 2                 |
| MTF2    | Q9Y483  | 0                 | 0              | 1                 |
| MYBL2   | P10244  | 5                 | 1              | 3                 |
| MYC     | P01106  | 135               | 30             | 19                |
| NASP    | P49321  | 1                 | 1              |                   |
| NCL     | P19338  | 24                | 2              | 3                 |
| NDC80   |         | 4                 | 2              | 0                 |
| NEK2    | P51955  | 4                 | 3              | 10                |
| NOLC1   | Q14978  | 2                 | 1              |                   |
| NOTCH2  | Q04721  | 13                | 2              | 9                 |
| NSD2    | O96028  | 2                 | 3              | 5                 |
| NUMA1   | Q14980  | 2                 | 1              | 2                 |
| NUP50   | Q9UKX7  | 1                 | 1              | 30                |
| NUP98   | P52948  | 4                 | 0              | 37                |
| NUSAP1  | Q9BXS6  | 3                 | 0              |                   |

| Gene     | UniProt | EuropePMC-IC hits | ChEMBL targets | Reactome pathways |
|----------|---------|-------------------|----------------|-------------------|
| ODC1     | P11926  | 0                 | 4              | 2                 |
| ODF2     | Q5BJF6  | 1                 | 0              | 7                 |
| ORC5     | O43913  | 0                 | 1              | 7                 |
| ORC6     | Q9Y5N6  | 0                 | 0              | 7                 |
| PAFAH1B1 | P43034  | 2                 | 0              | 14                |
| PBK      | Q96KB5  | 3                 | 1              |                   |
| PDS5B    | Q9NTI5  | 0                 | 1              | 4                 |
| PLK1     | P53350  | 6                 | 8              | 25                |
| PLK4     | O00444  | 4                 | 2              | 7                 |
| PML      | P29590  | 24                | 2              | 9                 |
| POLA2    | Q14181  | 1                 | 2              | 9                 |
| POLE     | Q07864  | 157               | 5              | 10                |
| POLQ     | O75417  | 0                 | 1              | 1                 |
| PRC1     |         | 6                 | 1              | 0                 |
| PRIM2    | P49643  | 1                 | 1              | 9                 |
| PRMT5    | O14744  | 5                 | 5              | 3                 |
| PRPF4B   |         | 0                 | 0              | 0                 |
| PTTG1    | O95997  | 3                 | 0              | 3                 |
| PTTG3P   | Q9NZH4  | 1                 | 0              |                   |
| PURA     | Q00577  | 8                 | 1              |                   |
| RACGAP1  | Q9H0H5  | 0                 | 1              | 12                |
| RAD21    | O60216  | 5                 | 0              | 7                 |
| RAD23B   | P54727  | 1                 | 1              | 4                 |
| RAD54L   | Q92698  | 0                 | 1              |                   |
| RASAL2   | Q9UJF2  | 0                 | 0              | 3                 |
| RBL1     | P28749  | 3                 | 1              | 7                 |
| RBM14    | Q96PK6  | 0                 | 1              | 1                 |
| RPA2     | P15927  | 3                 | 1              | 29                |
| RPS6KA5  | O75582  | 1                 | 2              | 5                 |
| SAP30    | O75446  | 0                 | 1              | 3                 |
| SFPQ     | P23246  | 3                 | 1              | 2                 |
| SLC12A2  | P55011  | 3                 | 2              | 1                 |
| SLC38A1  | Q9H2H9  | 3                 | 1              | 2                 |
| SLC7A1   | P30825  | 2                 | 1              | 1                 |
| SLC7A5   | Q01650  | 6                 | 3              | 3                 |
| SMAD3    | P84022  | 48                | 8              | 22                |
| SMARCC1  | Q92922  | 2                 | 1              | 11                |
| SMC1A    | Q14683  | 0                 | 1              | 7                 |
| SMC2     | O95347  | 1                 | 1              | 2                 |
| SMC4     | Q9NTJ3  | 0                 | 1              | 2                 |
| SNRPD1   | P62316  | 0                 | 2              | 7                 |
| SQLE     | Q14534  | 3                 | 3              | 2                 |
| SRSF1    | Q07955  | 9                 | 1              | 7                 |
| SRSF10   | O75494  | 0                 | 0              | 3                 |
| SRSF2    | Q01130  | 2                 | 1              | 7                 |
| SS18     | Q15532  | 5                 | 1              | 8                 |
| STAG1    | Q969W9  | 0                 | 0              | 1                 |
| STIL     | Q15468  | 1                 | 0              |                   |
| STMN1    | P16949  | 4                 | 1              | 1                 |
| SUV39H1  | O43463  | 4                 | 1              | 2                 |
| SYNCRIP  | O60506  | 2                 | 1              |                   |

| Gene   | UniProt | EuropePMC-IC hits | ChEMBL targets | Reactome pathways |
|--------|---------|-------------------|----------------|-------------------|
| TACC3  | Q9Y6A5  | 2                 | 2              | 2                 |
| TENT4A | Q5XG87  | 0                 | 1              | 2                 |
| TFDP1  | Q14186  | 0                 | 1              | 19                |
| TGFB1  | P01137  | 42                | 7              | 21                |
| TLE3   | Q04726  | 2                 | 0              | 5                 |
| TMPO   | P42166  | 0                 | 3              |                   |
| TNPO2  | O14787  | 0                 | 0              |                   |
| TOP1   | P11387  | 5                 | 8              | 1                 |
| TOP2A  | P11388  | 11                | 6              | 2                 |
| TPX2   | Q9ULW0  | 2                 | 2              | 2                 |
| TRA2B  | P62995  | 1                 | 1              | 5                 |
| TRAIP  | Q9BWF2  | 0                 | 0              | 1                 |
| TROAP  | Q12815  | 2                 | 0              |                   |
| TTK    |         | 2                 | 3              | 0                 |
| UBE2C  | O00762  | 2                 | 1              | 19                |
| UBE2S  | Q16763  | 0                 | 2              | 19                |
| UCK2   | Q9BZX2  | 1                 | 3              | 1                 |
| UPF1   | Q92900  | 2                 | 1              | 2                 |
| WRN    | Q14191  | 3                 | 1              | 17                |
| XPO1   | O14980  | 5                 | 8              | 19                |
| YTHDC1 | Q96MU7  | 2                 | 1              | 1                 |